Repetitionless

The offline documentation can be hard to follow so using the online documentation is recommended which you can find at https://docs.wilschack.dev/repetitionless. You can still use this if you need though, it has all the same information

William Schack

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1. Introduction

1.1 Features Video

https://youtu.be/BqwM6js7TkU

1.2 Description

Repetitionless is a set of shaders that removes texture tiling using multiple different toggleable techniques including:

- Voronoi noise and cell edge sampling to split up the texture and smooth between cells
- Random scaling and rotation between cells
- Triplanar sampling
- Variation texture support to add variation to the material
- Distance blending to either change the tiling and offset or material entirely at a set distance range
- Material Blending to overlay a separate material ontop of the main one by different noise functions or a custom mask

1.3 Contents

2. Installation

2.1 Download

Requirements:

- Unity 2021.3+

Repetitionless can be downloaded from:

- Unity Asset Store
- Itch.io

2.2 Installation

2.2.1 Unity Asset Store

Install it the same you would any other unity package:

1. Open the Unity Package Manager
2. Select Repetitionless under My Assets
3. Click Download, then Import after it is finished

2.2.2 Itch.io

You can install the package by either:

- Dragging the .unitypackage file into the project window
- Importing it through `Assets > Import Package > Custom Package`

3. Getting Started With Repetitionless

3.1 Tutorial Video

https://youtu.be/Ujk6ZyCmWak

3.2 Welcome Screen

When first importing the asset you will be greeted with a welcome screen with buttons to:

- View Documentation
- View and Reporting issues
- Import samples

To open this window again, you can find it in the toolbar at
`Windows > Repetitionless > Open Window`

3.3 Sample Scenes

The asset comes with 3 example scenes that you can view to see the different features of repetitionless and how to implement the materials

For instructions on accessing the samples:

View the Samples page

3.4 Using A Repetitionless Material

There are details you need to know when using the materials:

View the Material Details Page

For instructions on converting regular lit materials to repetitionless materials:

View the Converting Materials Page

For instructions on how to use the regular repetitionless shader:

View the Using Regular Materials Page

For instructions on how to use the layered repetitionless shader or a terrain:

View the Using Layered Materials Page

To view what each material property does:

View the Material Properties Page

3.5 Creating Shaders

You can view the Sub Graphs and Shader API documentation in the contents on the left of the page

If you want to create a shader graph or code shader using the features from the repetitionless materials:

View the Shader Creation page

3.6 Creating Scripts

You can view the Sub Graphs and Shader API documentation in the contents on the left of the page

If you want to create scripts or inspectors using the features from the repetitionless materials inspectors:

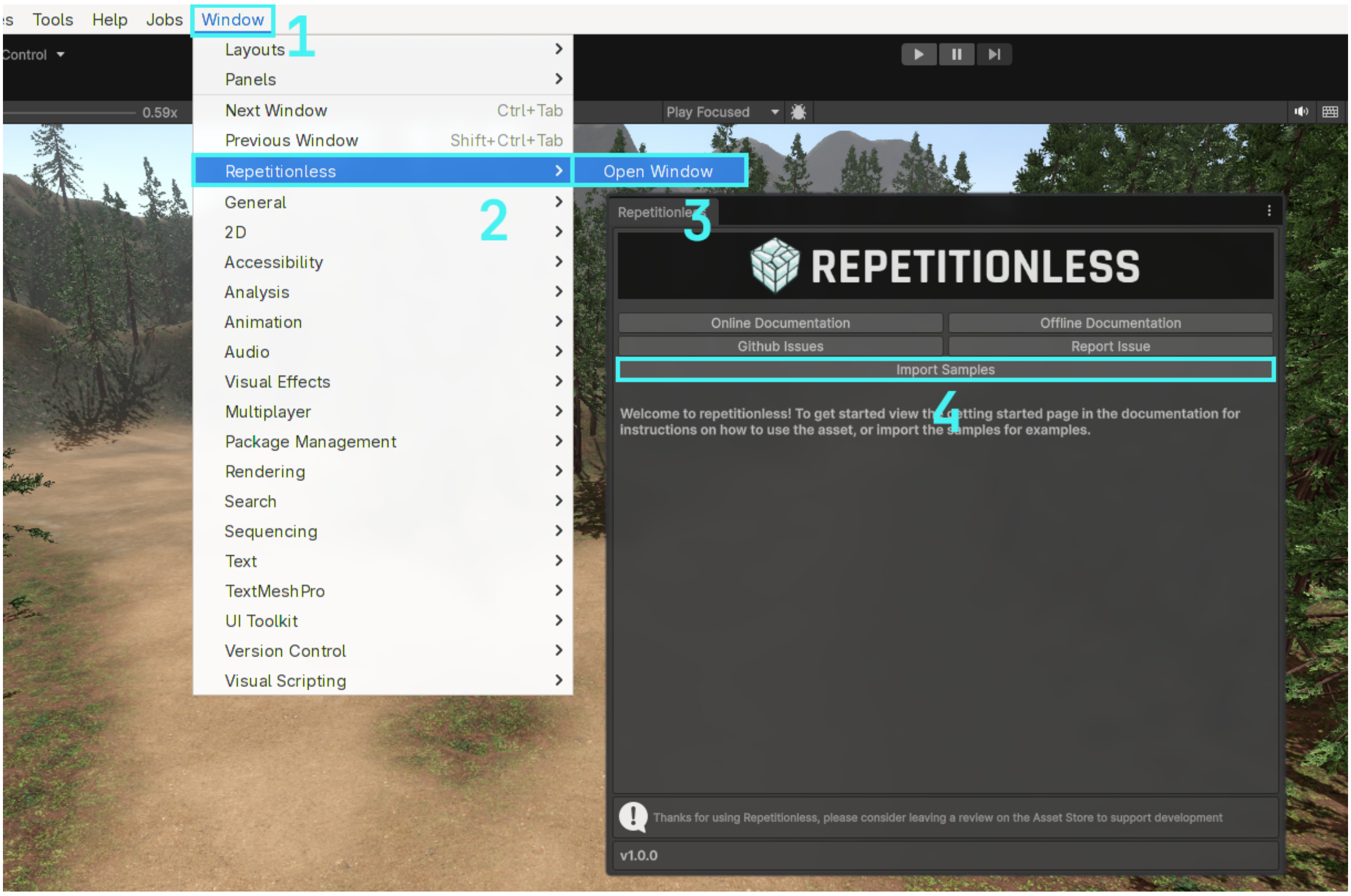
View the Scripting page

4. Samples

Samples are located in a folder which is hidden by default. To import the samples:

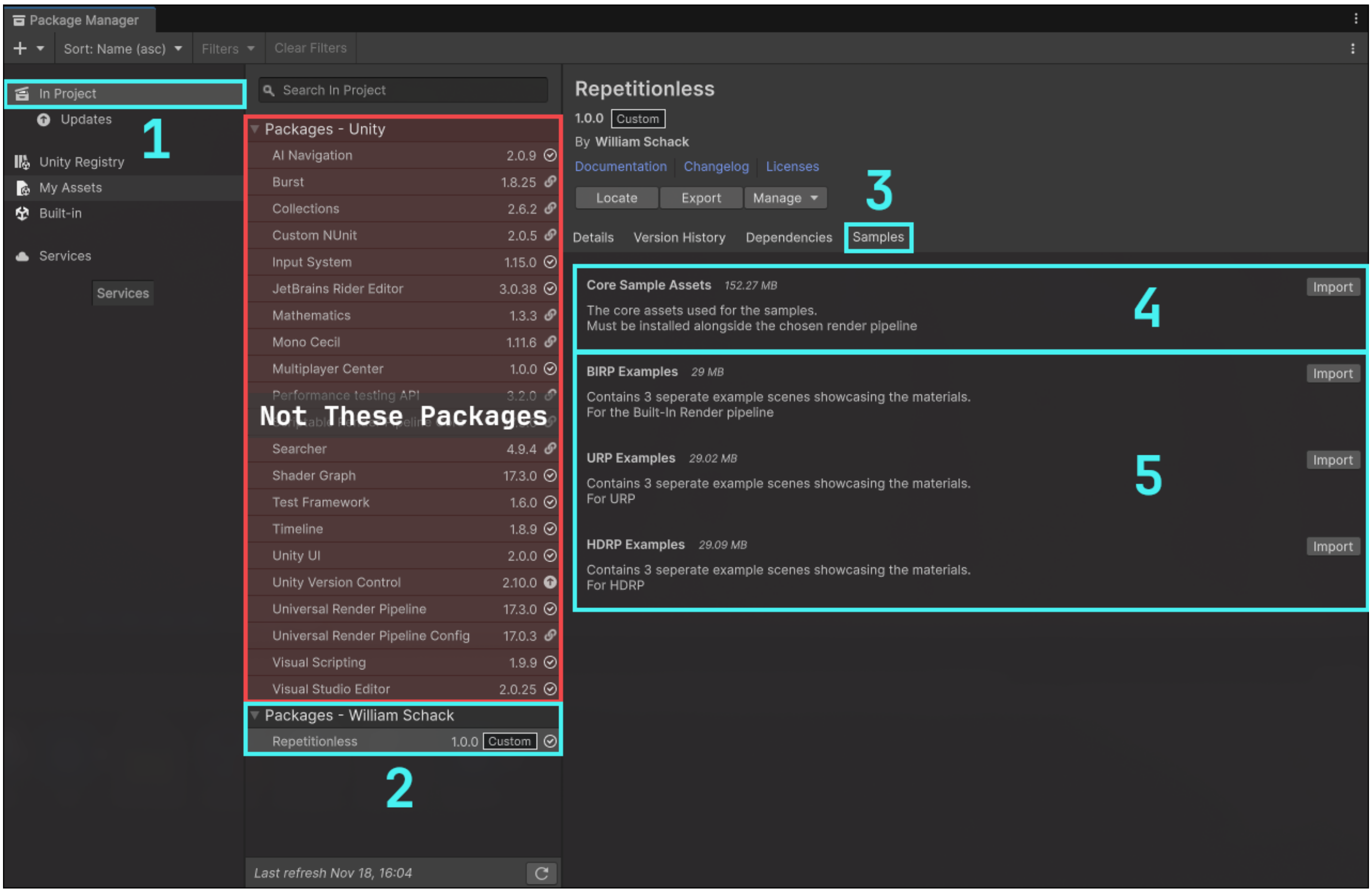
4.1 Automatic Installation

1. Open the windows tab in the toolbar
2. Navigate to `Repetitionless`
3. Click `Open Window`
4. Click the import samples button and it will automatically import your current render pipelines samples



4.2 Manual Installation

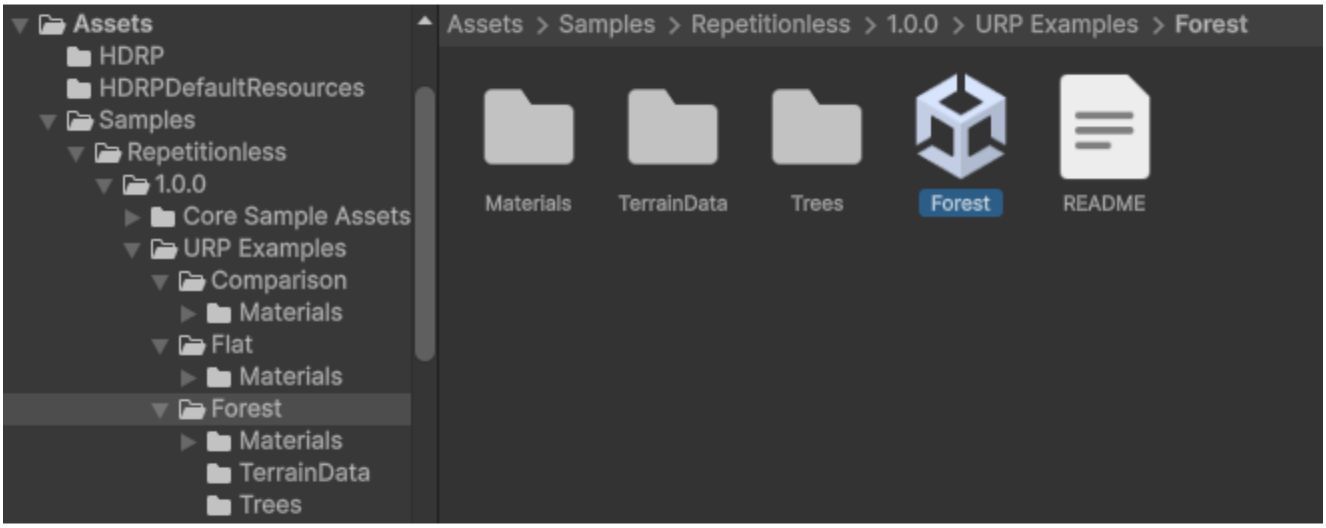
1. Open the Unity Package Manager and navigate the packages in the project
2. Select Repetitionless under `In Project > Packages - William Schack` *(Not `Packages - Asset Store`)*
3. Go to the samples tab
4. Click import on the `Core Sample Assets` before your render pipeline
5. Click import on the examples for your current render pipeline (`<RP> Examples`)



4.3 Opening The Samples

Samples will now be located in
`Assets/Samples/Repetitionless/<CurrentVersion>/<RP> Examples`

To open the sample scenes you can go to their respective folders and open the scene

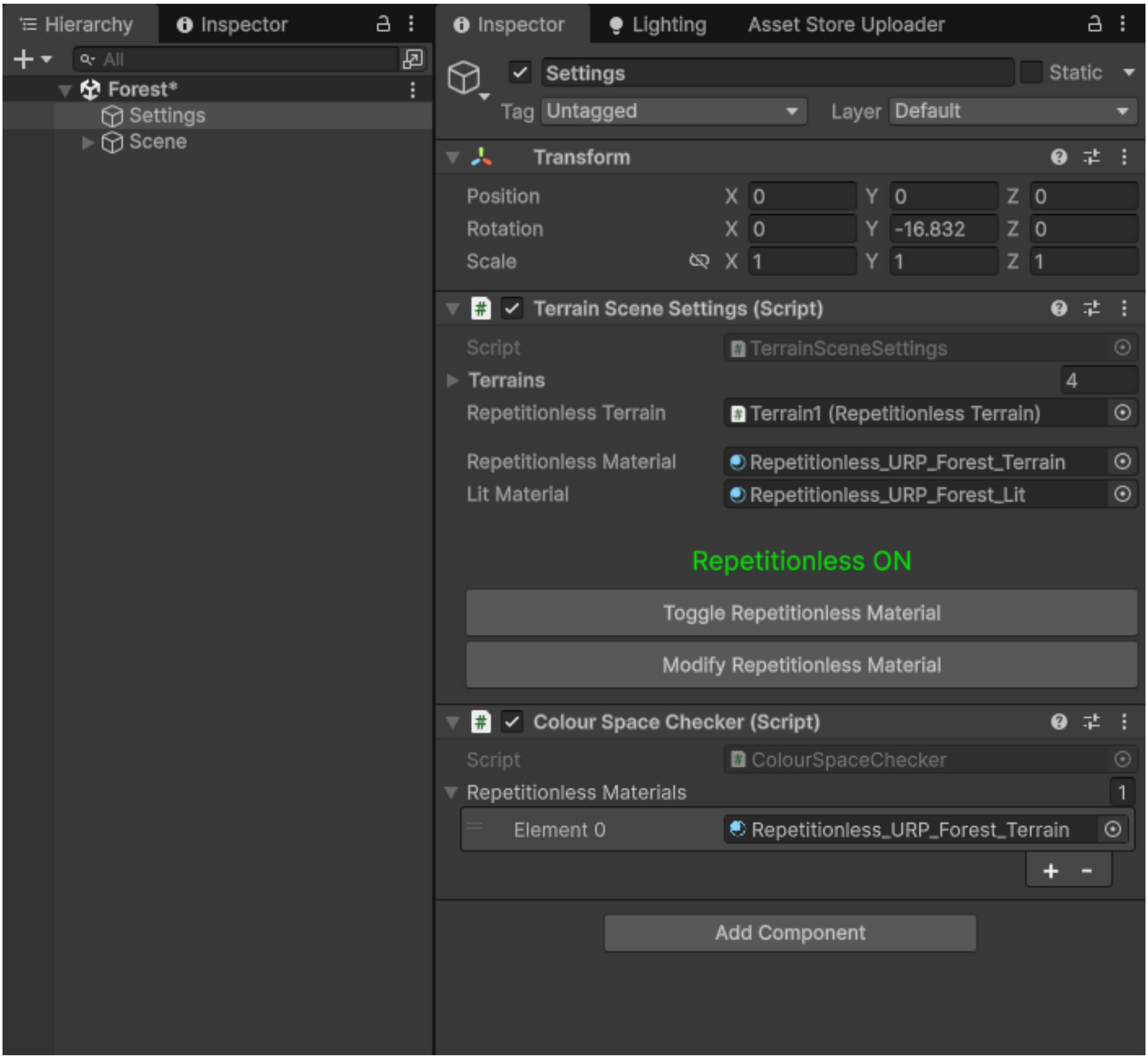


The Core Sample Assets folder stores the base assets for all the samples, this must be imported for the samples to work

4.4 Sample Settings

The samples have a settings object that has buttons to:

- Toggle between the repetitionless and regular lit materials
- Modify and view the material settings used



(The Colour Space Checker is used to make sure the textures are packed in the correct colour space, it is not needed after the scene is opened for the first time)

5. Material Details

5.1 Data Folder

All the properties and textures are stored in a folder along side the material named `<Material Name>_RepetitionlessData`

This folder is automatically handled and you do not need to worry about it

This folder will automatically be moved and deleted with the material but will not be copied so if you need to copy a material you will need to manually copy the folder aswell

5.2 Texture Packing

All the textures are packed into texture arrays for less samples in the shader which are stored in the data folder along side the material. These can be modified by clicking the **Array Settings button** in the Material Properties section

Arrays are split into:

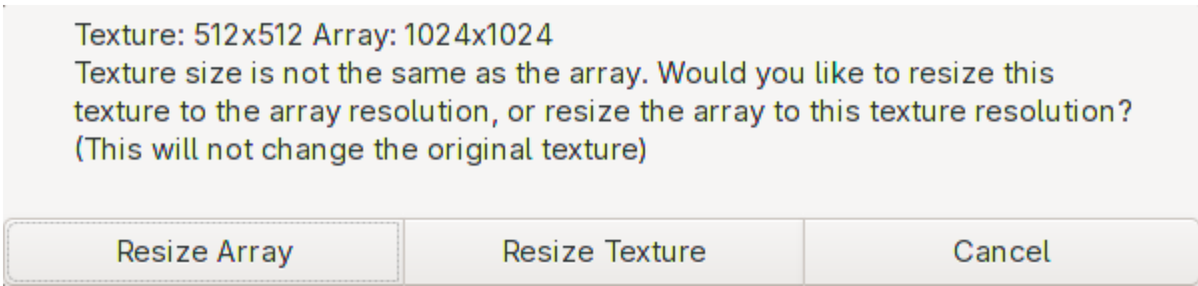
- AVTextures: Albedo (rgb), Variation (a)
- NSOTextures: Normal (rg), Smoothness (b), Occlussion (a)
- EMTextures: Emission (rgb), Metallic (a)
- BMTextures: Blend Mask (r)

Only the required arrays are created, and only the used textures are stored

The downside to texture arrays is that they can only have one resolution each array of textures, but this is automatically handled. This is also per array so for example the albedo textures can be a different resolution to the normal textures

When you assign a texture that is not the same resolution as the array, there is a popup that lets you either:

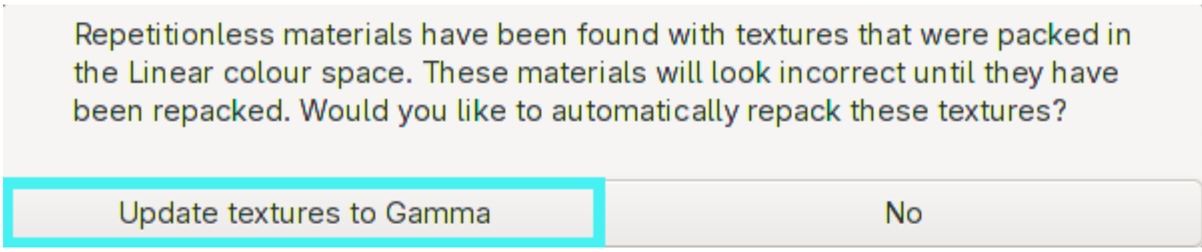
- Resize all the textures in the array to the resolution of the new texture
- Resize the texture to the same resolution as the array



5.3 Colour Space

sRGB textures (Albedo/Colour) are packed with the set colour space in mind. They will need to be repacked when changing the colour space

There is a popup when changing the colour space if any textures need to be repacked that will automatically repack the textures for you.
Make sure to select **Update Textures to \<Colour Space>** for the textures to look correct



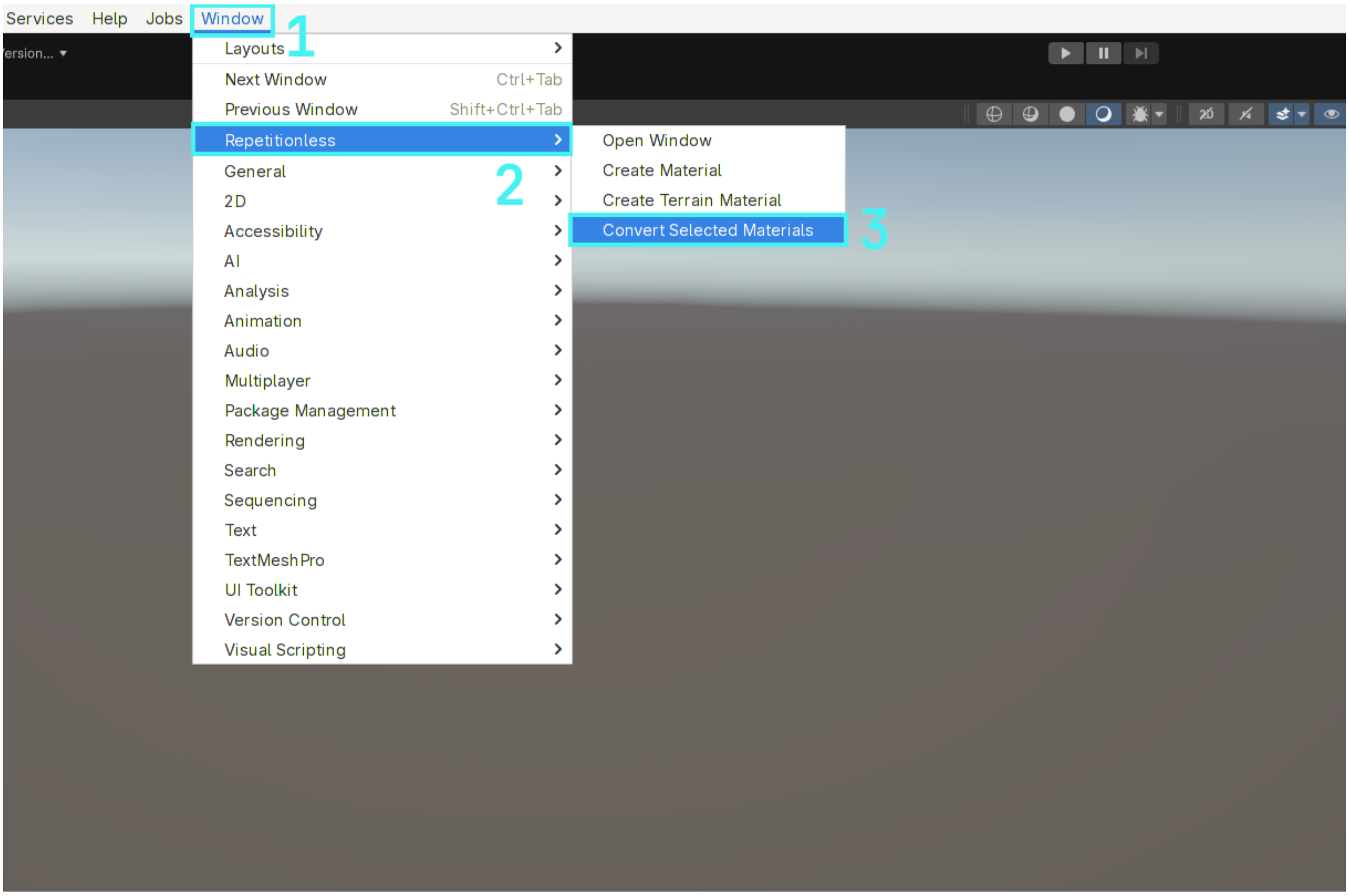
To view what each property does, visit the Material Properties page

6. Converting Materials

6.1 Converting A Material

To convert a regular lit material to a repetitionless material:

1. Select the materials you want to convert
2. Open the windows tab in the toolbar
3. Navigate to `Repetitionless`
4. Click `Convert Selected Materials`
5. The materials will be created next to the original materials with the suffix "_repetitionless"



Important Details:

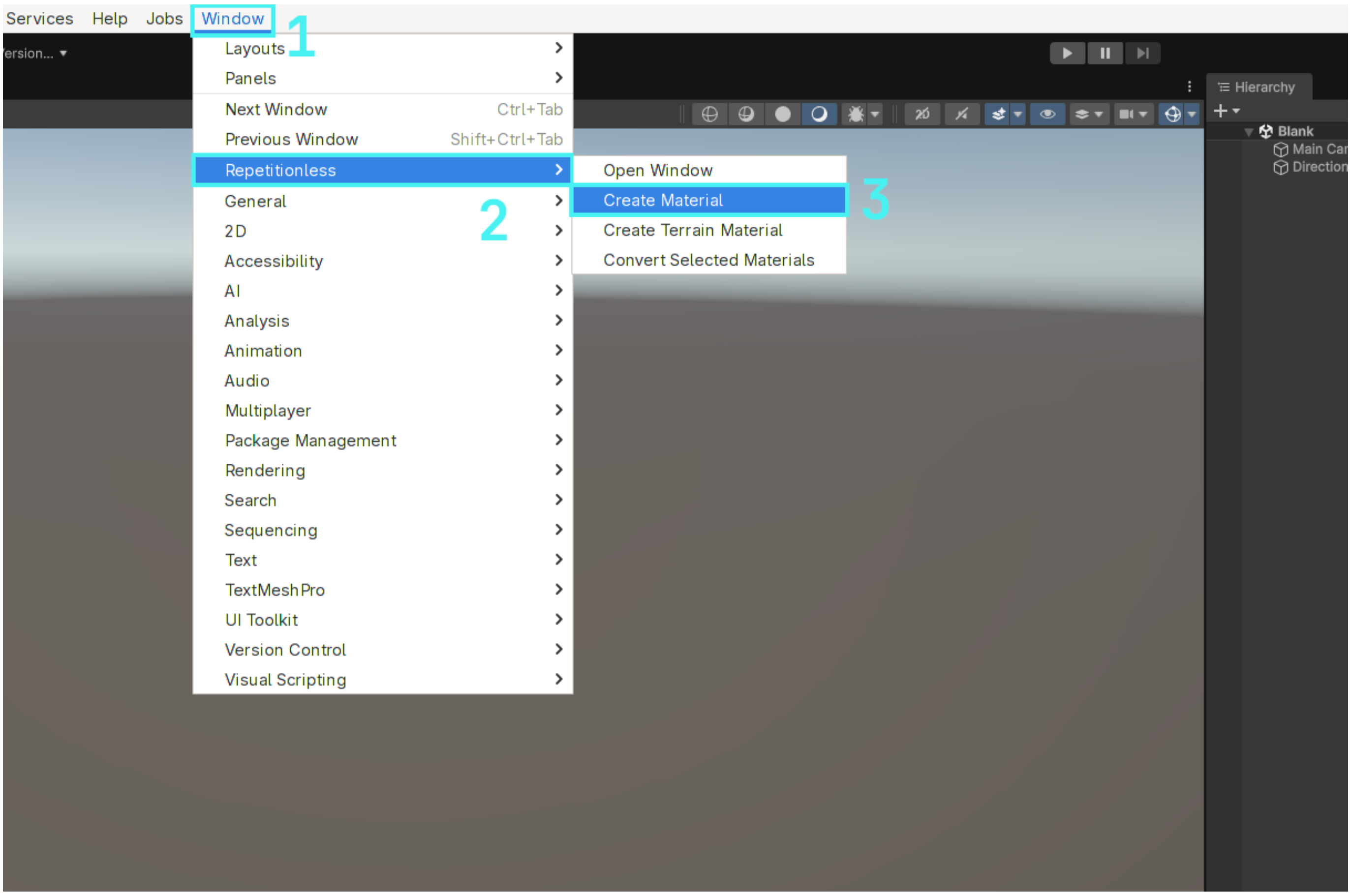
- This will always convert to the same render pipeline as the input material
- This works for BIRP, URP, & HDRP standard lit materials, any others will not work

7. Using Regular Materials

7.1 Creating A Material

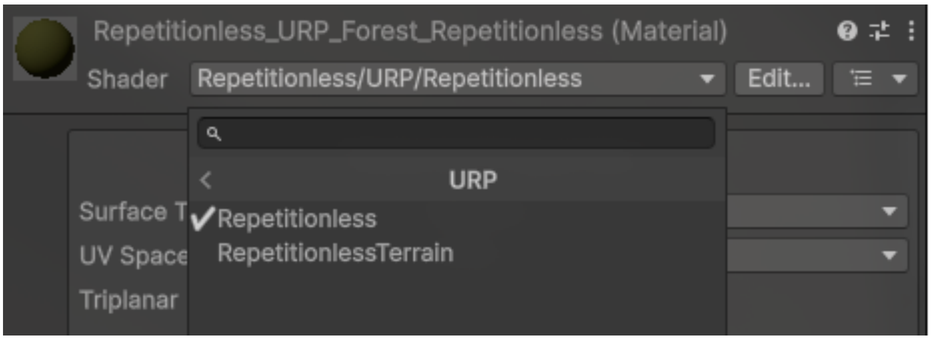
7.1.1 Automatic

1. Open the windows tab in the toolbar
2. Navigate to `Repetitionless`
3. Click `Create Material`
4. The material will be created in the current folder in the project window



7.1.2 Manual

1. Create a material
2. Select the shader dropdown
3. Navigate to `Repetitionless`
4. Select your render pipeline
5. Select `Repetitionless`

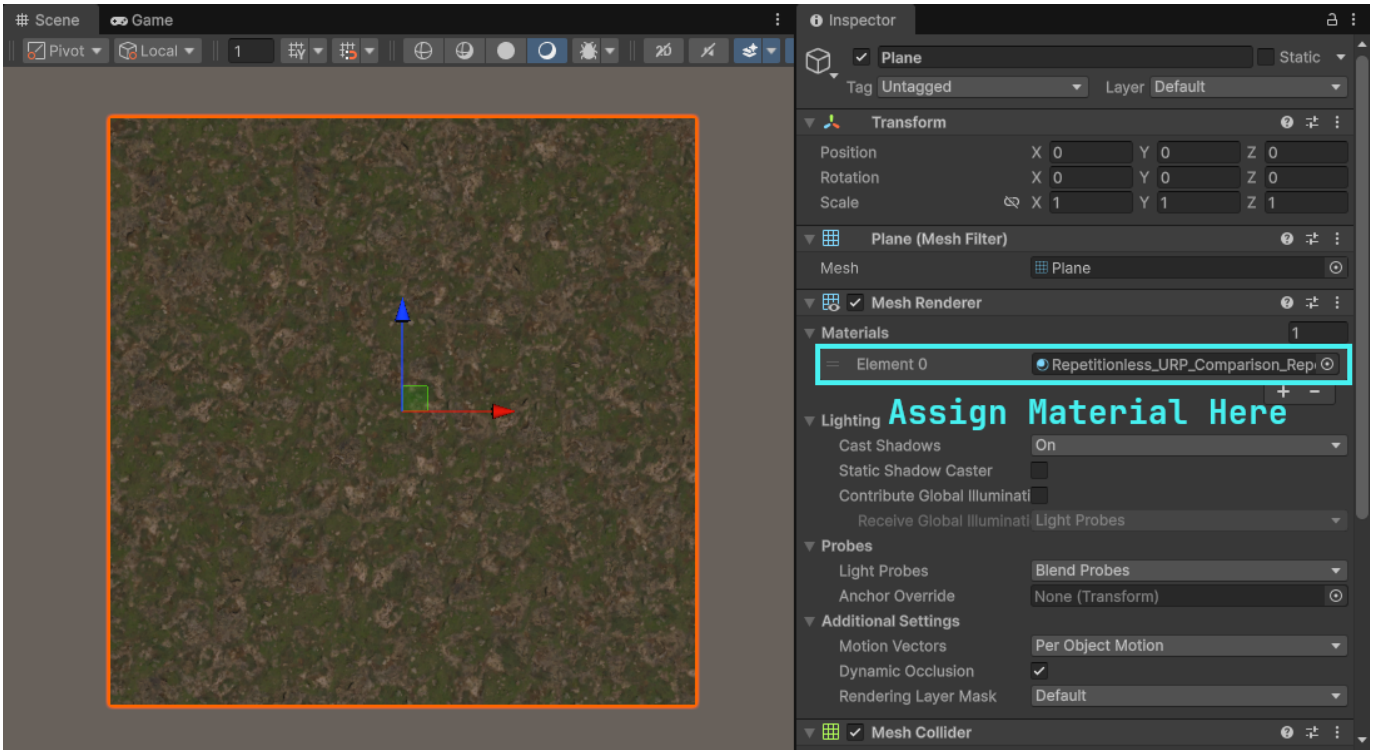


Important Details:

- Render pipelines can be switched without losing data. ex. Changing a Repetitionless material from BIRP to its URP version will keep all the settings
- Material data is stored in a folder along side the material named "{MaterialName}_Data". This is automatically moved and deleted with the material but it will need to be manually copied when copying the material

7.2 Using The Material

You can just assign repetitionless material to any Mesh Renderer and it will be applied to that object



7.3 Configuration

All the materials can be configured through its inspector by selecting the material and viewing the inspector window

To view what each property does, visit the Material Properties page

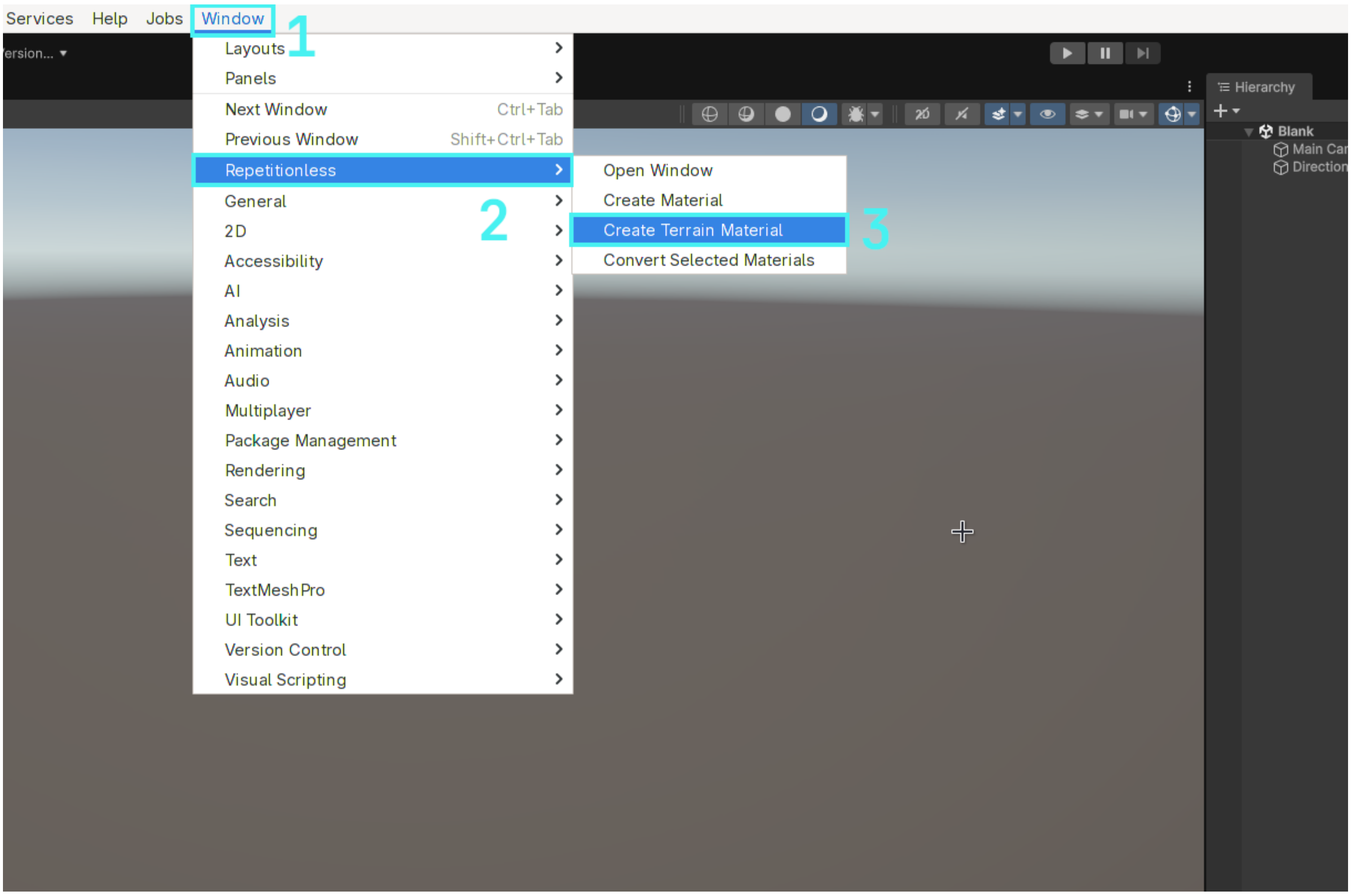
8. Using Layered Materials

The layered material is used to display up to 32 different material layers on one material. Layer positions can either be updated with control textures or synced with a terrain. Instructions are below for both

8.1 Creating A Layered Material

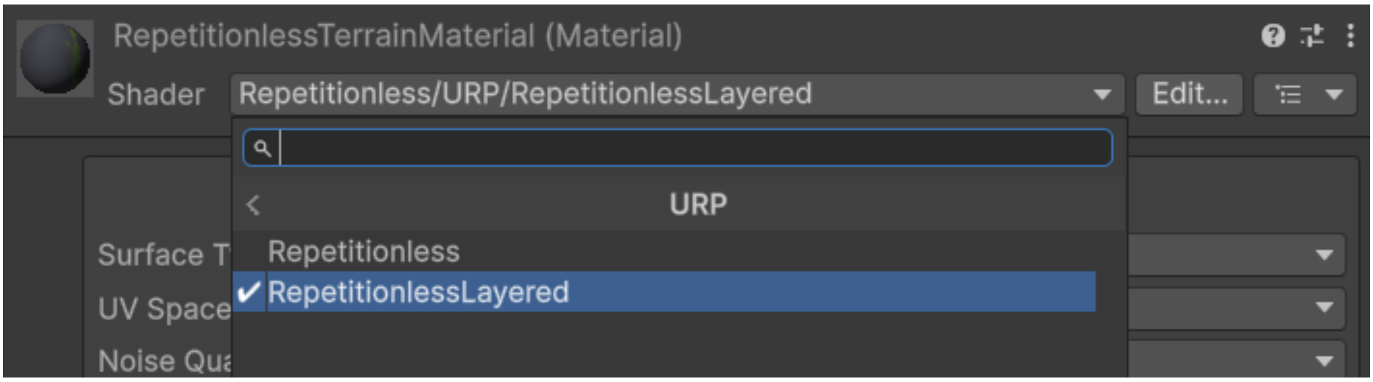
8.1.1 Automatic

1. Open the windows tab in the toolbar
2. Navigate to `Repetitionless`
3. Click `Create Layered Material`
4. The material will be created in the current folder in the project window



8.1.2 Manual

1. Create a material
2. Select the shader dropdown
3. Navigate to `Repetitionless`
4. Select your render pipeline
5. Select `RepetitionlessLayeredTerrain` or `RepetitionlessLayeredLit`



Important Details:

- Render pipelines can be switched without losing data. ex. Changing a Repetitionless material from BIRP to its URP version will keep all the settings
- Material data is stored in a folder along side the material named "{MaterialName}_RepetitionlessData". This is automatically moved and deleted with the material but it will need to be manually copied when copying the material
- The `LayeredTerrain` shader is used when the mode is set to Terrain Layers, `LayeredLit` is used when the mode is set to anything else. This can be changed in the inspector, so when creating the material you can select either.

8.2 Using Terrain Layers

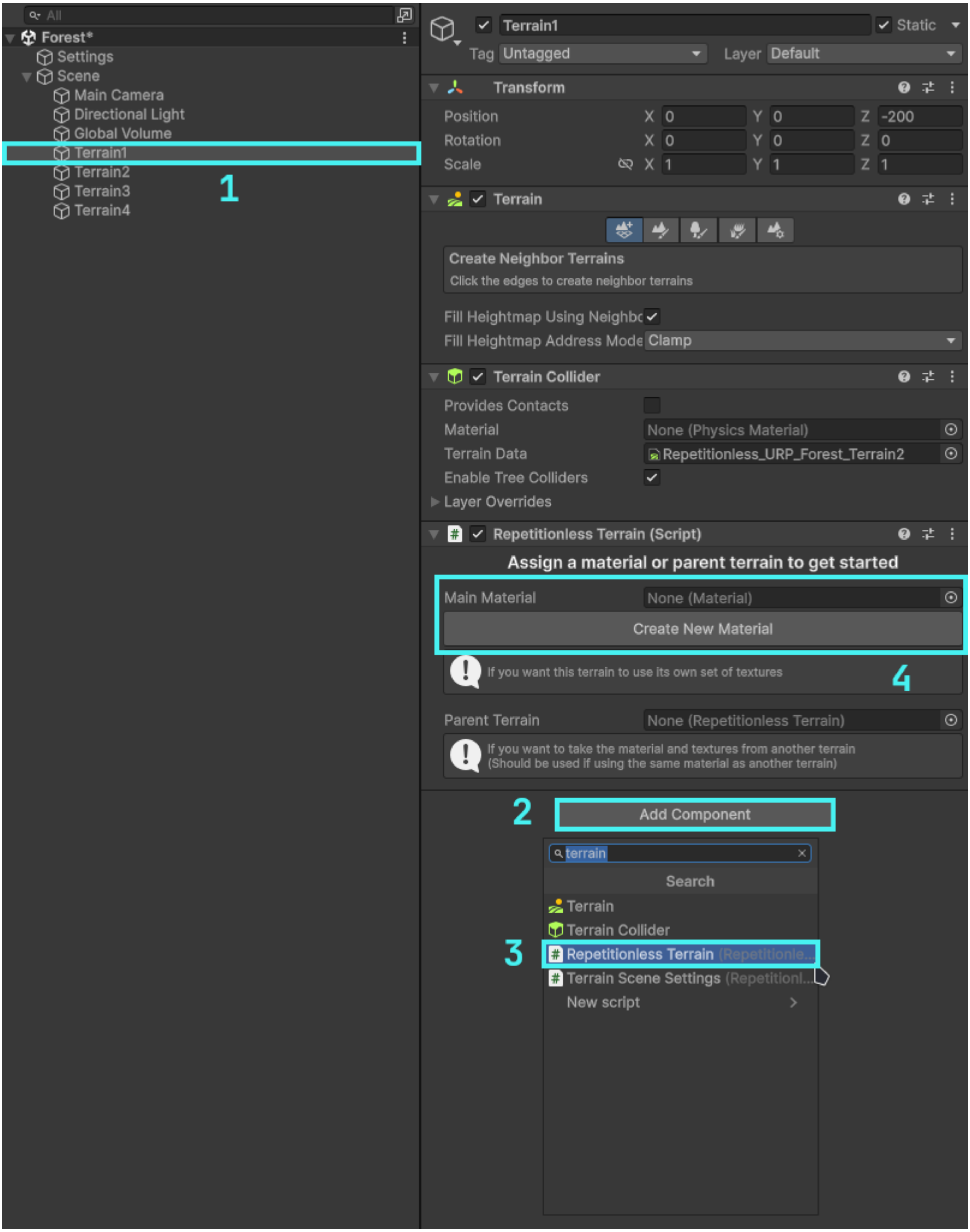
To use a terrain with the repetitionless material you have to add a `RepetitionlessTerrain` component to the terrain which can be done as follows:

1. Select the terrain you want to use
2. Select `Add Component`
3. Select `Repetitionless Terrain`

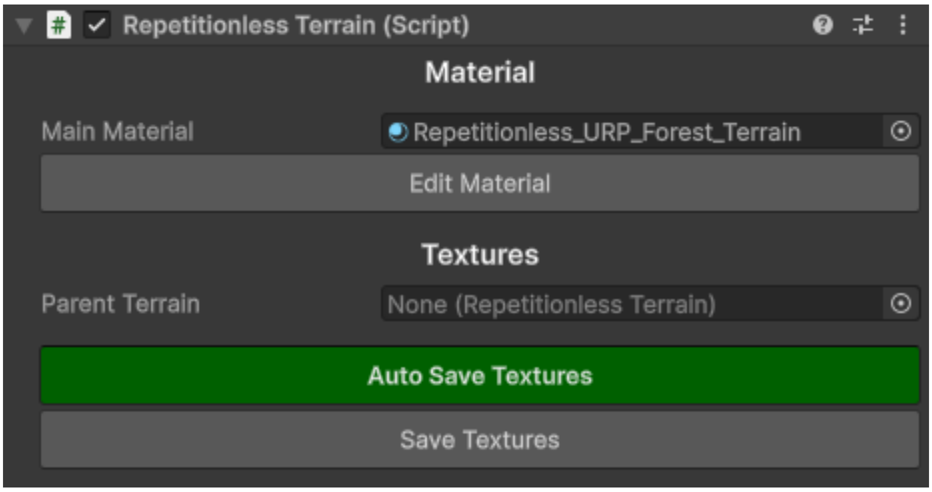
To add a material you can either:

- Click the `Create New Material` button
- Assign a material in the `Main Material` field

Note the the material field will only allow Repetitionless Layered materials. Assigning a material will also automatically set its layer mode to Terrain Layers


All
Forest*
Settings
Scene
Main Camera
Directional Light
Global Volume
Terrain1
Terrain2
Terrain3
Terrain4
1
Terrain1
Static
Tag Untagged
Layer Default
Transform
Position X 0 Y 0 Z -200
Rotation X 0 Y 0 Z 0
Scale X 1 Y 1 Z 1
Terrain
Create Neighbor Terrains
Click the edges to create neighbor terrains
Fill Heightmap Using Neighbo
Fill Heightmap Address Mode Clamp
Terrain Collider
Provides Contacts
Material None (Physics Material)
Terrain Data Repetitionless_URP_Forest_Terrain2
Enable Tree Colliders
Layer Overrides
Repetitionless Terrain (Script)
Assign a material or parent terrain to get started
Main Material None (Material)
Create New Material
If you want this terrain to use its own set of textures
4
Parent Terrain None (Repetitionless Terrain)
If you want to take the material and textures from another terrain
(Should be used if using the same material as another terrain)
2
Add Component
terrain
Search
Terrain
Terrain Collider
3
Repetitionless Terrain (Repetitionle...
Terrain Scene Settings (Repetitionl...
New script

8.2.1 Using The Terrain Script

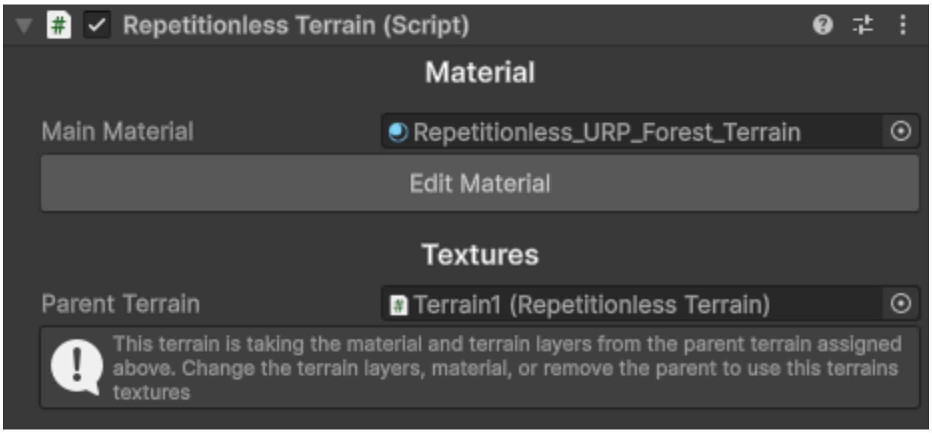


That is basically all you need to do! The script will handle the material assigning and texture syncing for you

With Auto Save Textures enabled, the terrain will sync whenever you update the terrain layers, updating the materials textures. With it disabled, it will not sync the textures until you click the Save Textures button.

If you ever have any errors with the terrain, first see if clicking the Save Textures button fixes your issue. If it doesnt feel free to submit a bug report

8.2.2 Using Multiple Terrains

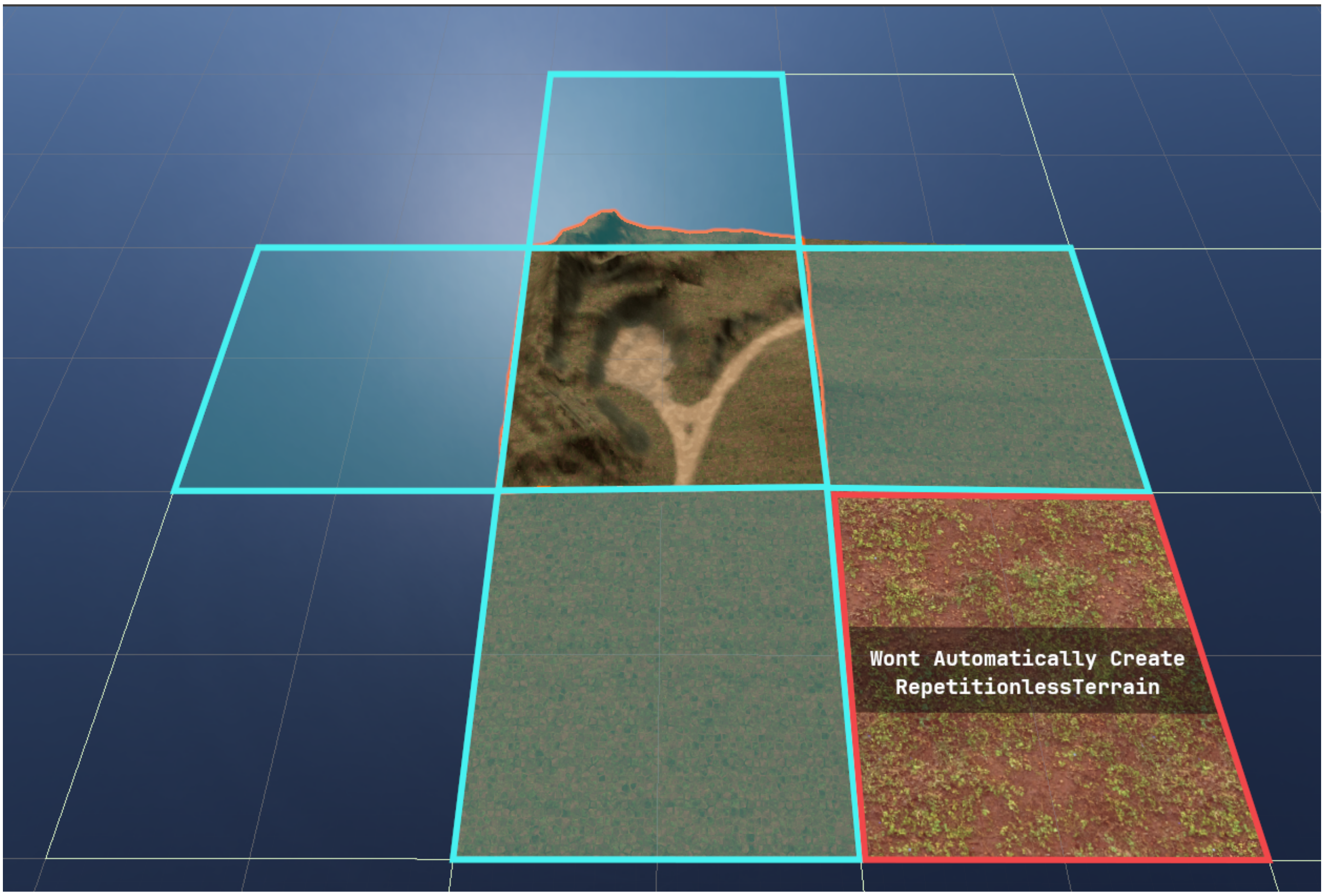


The script also has a Parent Terrain field that when set will update that terrain to use the same material and layers from the parent. Keep in mind that when setting a parent terrain it will overwrite the current terrain layers you have set.

If a terrain has a parent, it will automatically update when the parent terrain updates, so using the save textures button on the main terrain will update all the children aswell.

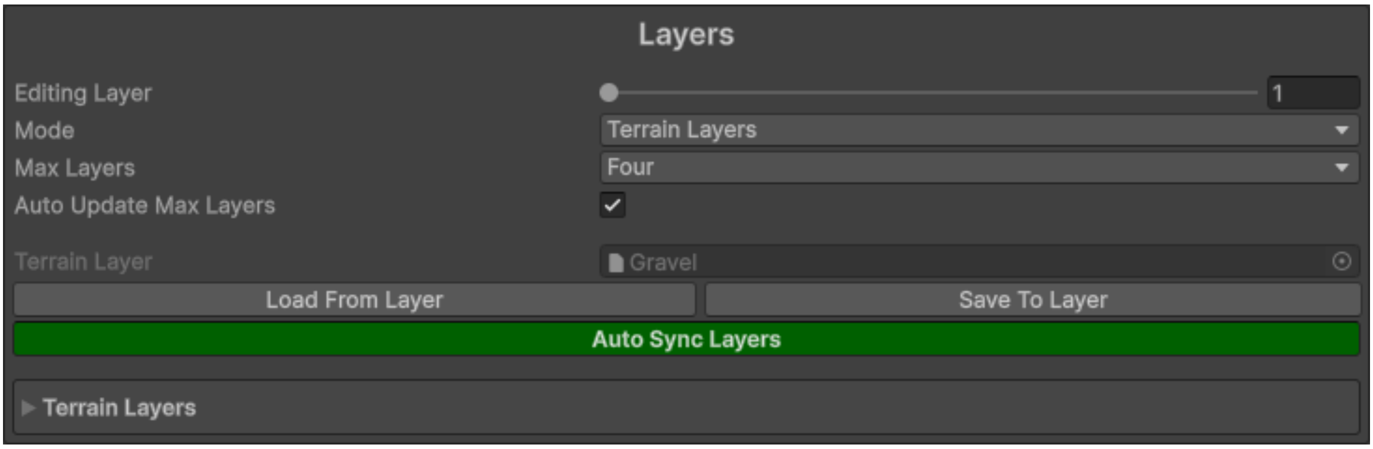
Important Details:

- One parent terrain can be assigned to as many child terrains as you like
- Terrain Materials should only be synced to one set of terrain layers, but you can use the parent terrain field to get around that
- The parent terrain will automatically detach when you either change the terrain layers, or update the material



On terrains with the RepetitionlessTerrain, when using the Create Neighbour Terrains tool and creating a directly adjacent terrain, it will automatically add a RepetitionlessTerrain component to that terrain with the selected one as the parent. This will not work for non-adjacent terrains though as shown in the image above

8.2.3 Editing The Terrain Layers



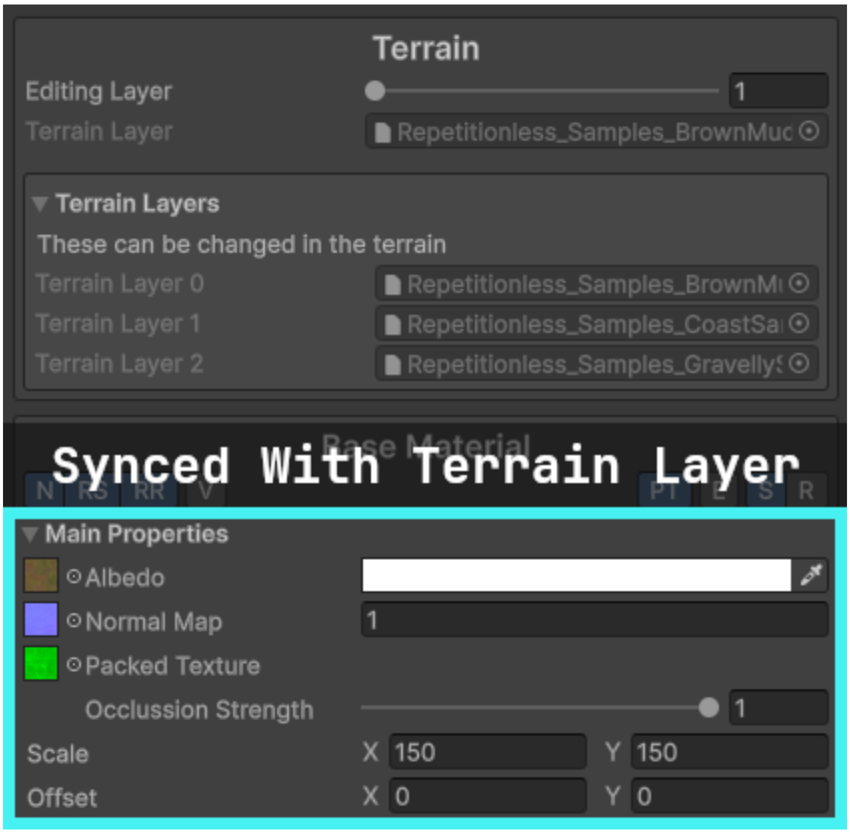
Setting	Description
Editing Layer	The layer currently being edited in the inspector
Mode	The mode of this material. This changes the shader depending on the option with Terrain Layers changing it to a terrain compatible shader
Max Layers	The max layers allowed on this material. Set lower when possible for performance improvements
Auto Update Max Layers	Automatically updates the above Max Layers to fit your assigned layer count

After modifying any field in the terrain layer and saving, it will automatically update those properties in any material that is linked to that layer if the auto sync is enabled. You can manually load or save to and from the layer with the buttons

The properties effected and what they link to include:

- Diffuse Map > Base Albedo Texture
- Normal Map > Base Normal Map
- Normal Scale > Base Normal Scale
- Mask Map > Base Packed Texture (Will enable packed texture when assigned)
- Metallic > Base Metallic
- Smoothness > Base Smoothness
- Tiling > Base Tiling
- Offset > Base Offset

8.2.4 Editing The Material



If a terrain has been assigned, when editing the material it will show the terrain layers that it is linked to and each layer can be edited individually.

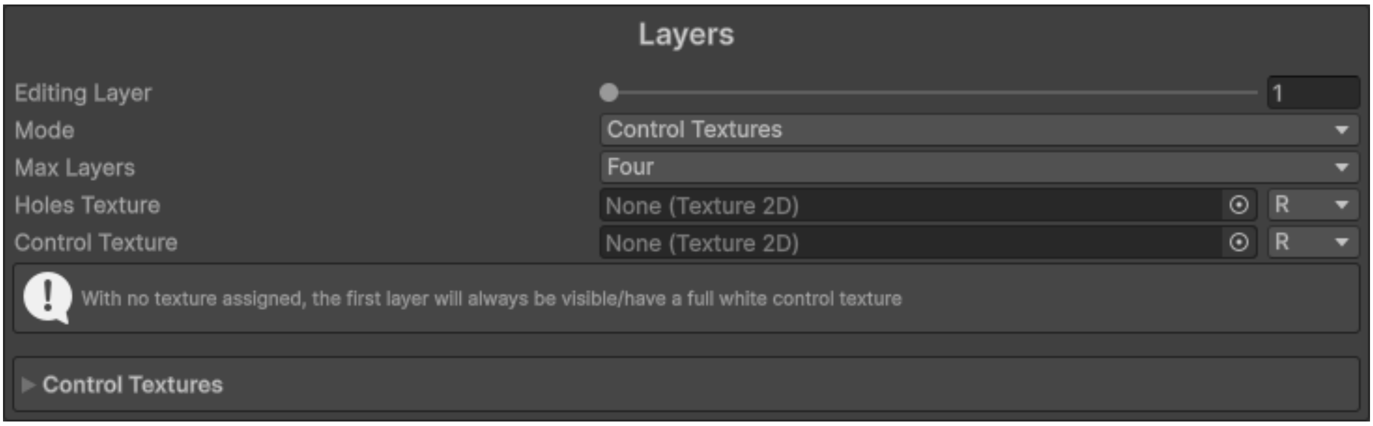
Updating the Main Properties in the Base Material that have an equivalent in the terrain layer (listed above) will also automatically update the terrain layer

Layers in the repetitionless material correspond to the order of the assigned terrain layers as shown in the image above

The shader supports up to 32 terrain layers. Any more wont work and will appear white

8.3 Using Control Textures

To use control textures, first make sure the layer mode is set to `Control Textures`



Setting	Description
Editing Layer	The layer currently being edited in the inspector
Mode	The mode of this material. This changes the shader depending on the option with Terrain Layers changing it to a terrain compatible shader
Max Layers	The max layers allowed on this material. Set lower when possible for performance improvements
Holes Texture	The holes texture that will be used for the material. It will read from the selected channel
Control Texture	The control texture that will be used for this layer. It will read from the selected channel

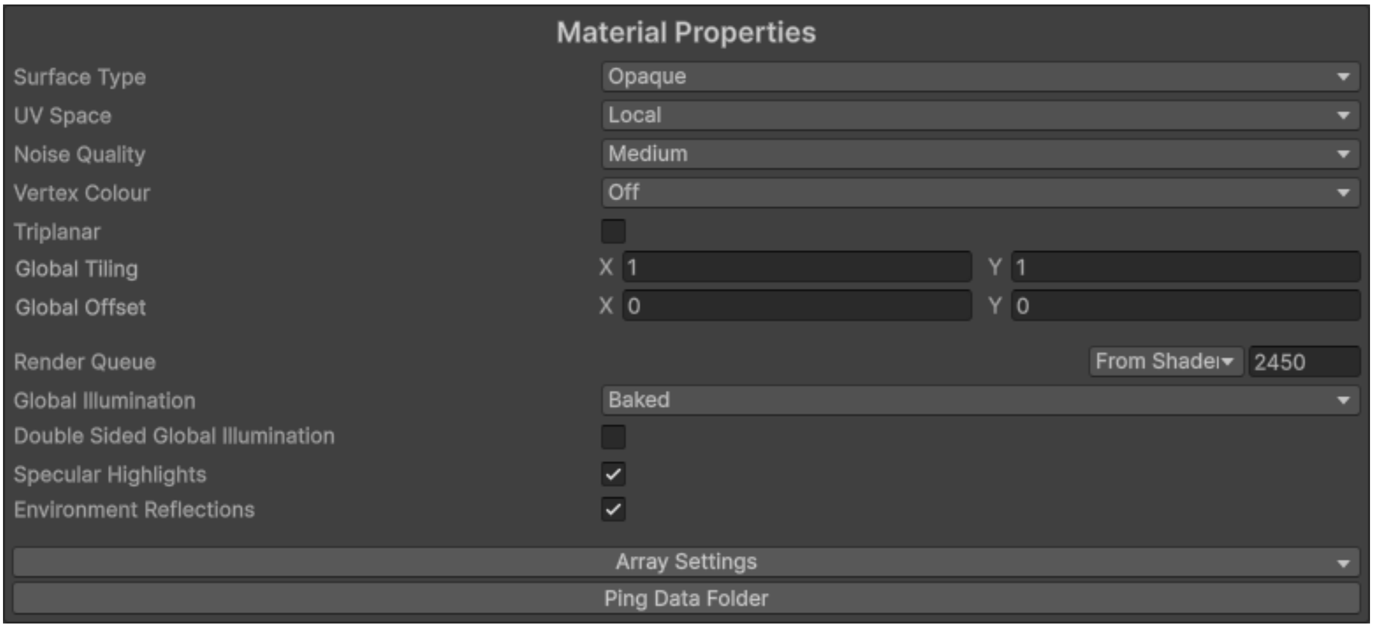
The control textures for a specific pixel must add up to 1. For example, if you want a spot to be the second layer, the second control texture will have a value of 1 in that spot where other layers will have a value of 0.

Example of two control textures where the left is layer 1, and the right layer 2

Everything else is the same as the regular repetitionless material. To view what each property does, visit the Material Properties page

9. Material Properties

9.1 Material Properties



Has all the general properties for the material

Property	Description
Surface Type	The surface type of the material 0: Opaque 1: Cutout 2: Transparent
UV Space	Sets the UV Space 0: Local 1: World *Note that with world space UVs you will require larger Tiling*
Noise Quality	How the noise is sampled High: Dynamically samples noise *(Required for non-repeating noise and the angle offset setting)* Medium: Uses the pre-rendered 4k texture Low: Uses the pre-rendered 1k texture
Vertex Colour	How the vertex colour is blended with the output colour 0: Off 1: Multiply 2: Additive 3: Subtractive 4: Overwrite
Triplanar	Enables triplanar sampling and forces world UVs *Note that the first time this is set for a material it will take some time to finish as it will recompile the shader, it will be instant for subsequent toggles*
Global Tiling	This is multiplied with each materials tiling
Global Offset	This is added onto each of the materials offset
Render Queue	Changes when the material is drawn
Global Illuminattion	Controls if the global illumination is baked or realtime 0: Realtime 1: Baked 2: None
Double Sided Global Illumination	If the lightmapper accounts for both sides of the geometry when calculating Global Illumination
Specular Highlights	When enabled, the Material reflects the shine from direct lighting
Environment Reflections	When enabled, the Material samples reflections from the nearest Reflection Probes or Lighting Probe
Array Settings	Shows a dropdown with buttons to modify each of the texture arrays that the textures are stored in AVTextures: Albedo (rgb), Variation (a) NSOTextures: Normal (rg), Smoothness (b), Occlussion (a) EMTextures: Emission (rgb), Metallic (a) BMTextures: Blend Mask (r)
Ping Data Folder	Pings the data folder

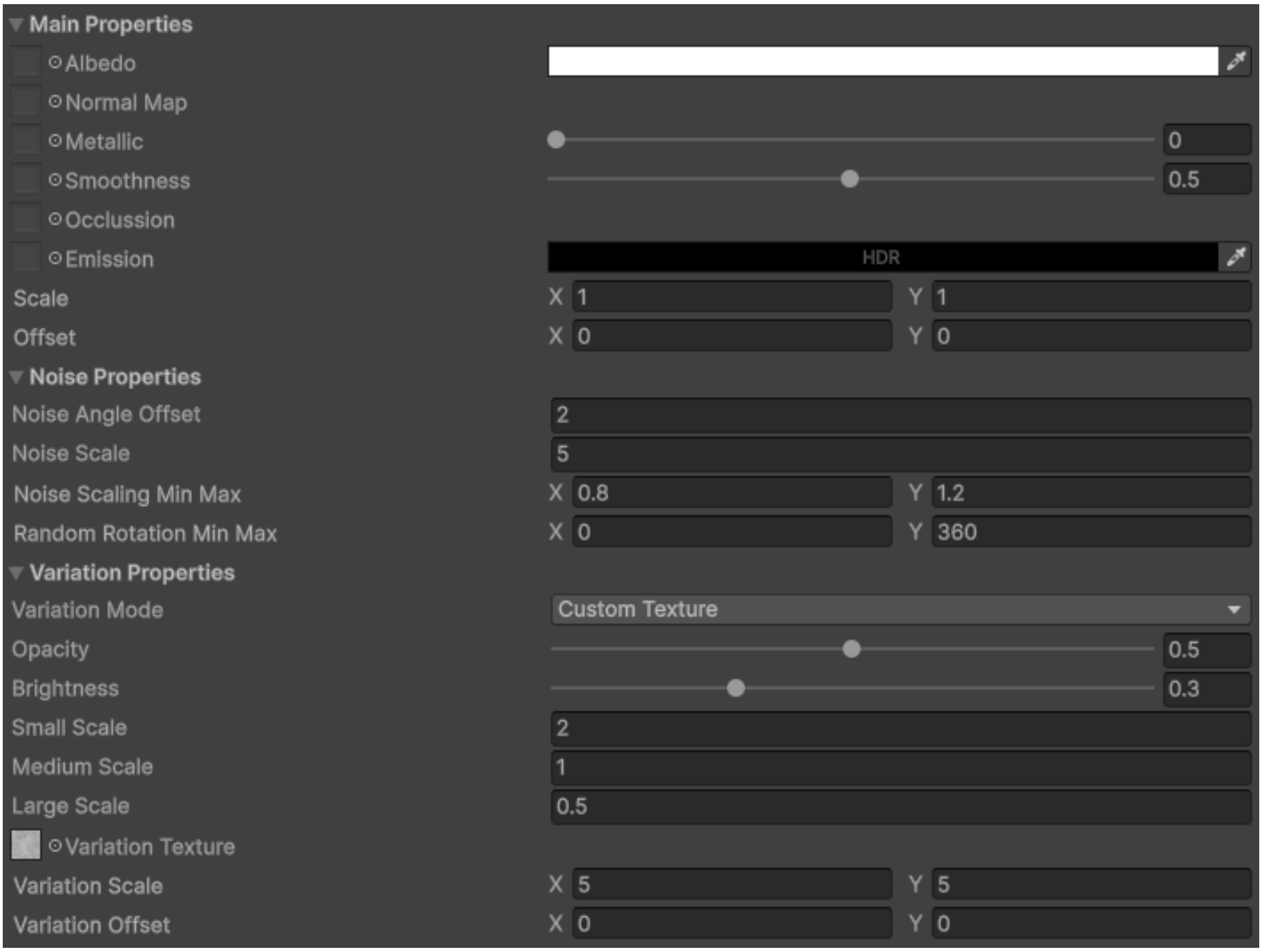
9.2 Toolbar

Each material has a toolbar with various settings

Each settings text will shrink to the first letters of each word if the inspector width is too small for the whole words to fit

Setting	Description
Noise	Adds random scaling & rotation based on voronoi noise
Random Scaling	Adds random scaling to each voronoi cell *Only shown if Noise is enabled*
Random Rotation	Adds random rotation to each voronoi cell *Only shown if Noise is enabled*
Variation	Adds random variation on top of the albedo color **Using a custom texture can cause visible tiling**
Packed Texture	If you are using a packed texture of multiple regular ones (Better for performance) R: Metallic G: Occlussion A: Smoothness/Roughness
Emission	If Emission is enabled
Smooth/Rough	Toggles between the smoothness and roughness texture **If you are using a packed texture it will sample it with a smoothness or roughness texture based on this toggle**

9.3 Material Settings



Contains all the settings for a material. Settings can be enabled and disabled through the toolbar

This is shown for each enabled material (Base material, distance blend material, blend material)

When a texture is assigned for fields that only use one texture channel, it will have a dropdown where you can choose which channel to use

9.3.1 Main Properties

Property	Description
Albedo	Albedo (RGB)
NormalMap	Normal map (RG)
Metallic	Metallic (R)
Smoothness/Roughness	Smoothness/Roughness (R) *Changes between smoothness and roughness based on toolbar setting*
Occlussion	Occlussion (R)
Emission	Emission (RGB)
Packed Texture	The mask map for the current material R: Metallic G: Occlussion A: Smoothness/Roughness *- Only shown if Packed Texture is enabled* *- Alpha toggles between smoothness and roughness based on toolbar setting*
Occlussion Strength	The occlussion strength of the packed texture occlussion *Only shown if Packed Texture is enabled*
Scale	The texture tiling
Offset	The texture offset

9.3.2 Noise Properties

These are only shown if Noise is enabled in the toolbar

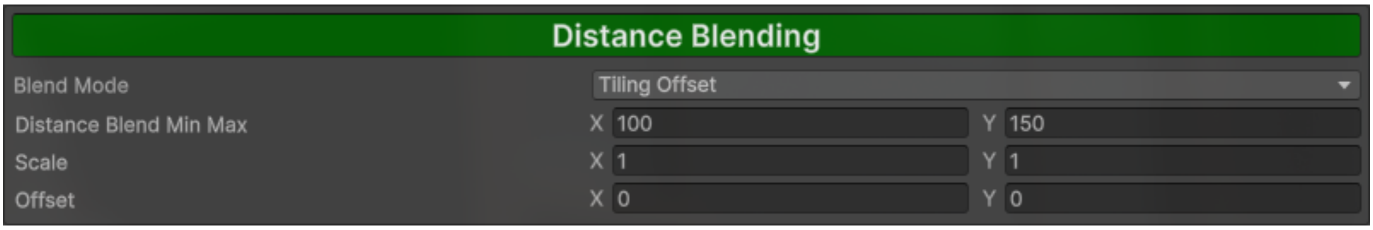
Property	Description
Noise Angle Offset	The angle offset of the voronoi noise *Only shown if noise quality is set to high*
Noise Scale	The scale of the voronoi noise
Noise Scaling Min Max	Range that each voronoi cell is randomly scaled by x: Min Scale y: Max Scale *Only shown if Random Scaling is enabled in the toolbar*
Random Rotation Min Max	Range that each voronoi cell is randomly rotated by x: Min Rotation Degrees y: Max Rotation Degrees *Only shown if Random Rotation is enabled in the toolbar*

9.3.3 Variation Properties

These are only shown if Variation is enabled in the toolbar

Property	Description
Variation Mode	The variation mode used Using a custom texture can cause visible tiling 0: Perlin noise 1: Simplex noise 2: Custom texture
Opacity	Transparency of the variation
Brightness	Intensity of the variation
Small Scale	Scale of the small variation sample
Medium Scale	Scale of the medium variation sample
Large Scale	Scale of the large variation sample
Noise Strength	Strength of the noise *Only shown if Variation Mode is set to any noise*
Noise Scale	The noise tiling *Only shown if Variation Mode is set to any noise*
Noise Offset	The noise offset *Only shown if Variation Mode is set to any noise*
Variation Texture	Texture that is drawn onto other materials, can cause visible tiling Variation (R), other channels are ignored *Only shown if Variation Mode is set to Custom Texture*
Variation Scale	The variation tiling *Only shown if Variation Mode is set to Custom Texture*
Variation Offset	The variation offset *Only shown if Variation Mode is set to Custom Texture*

9.4 Distance Blending

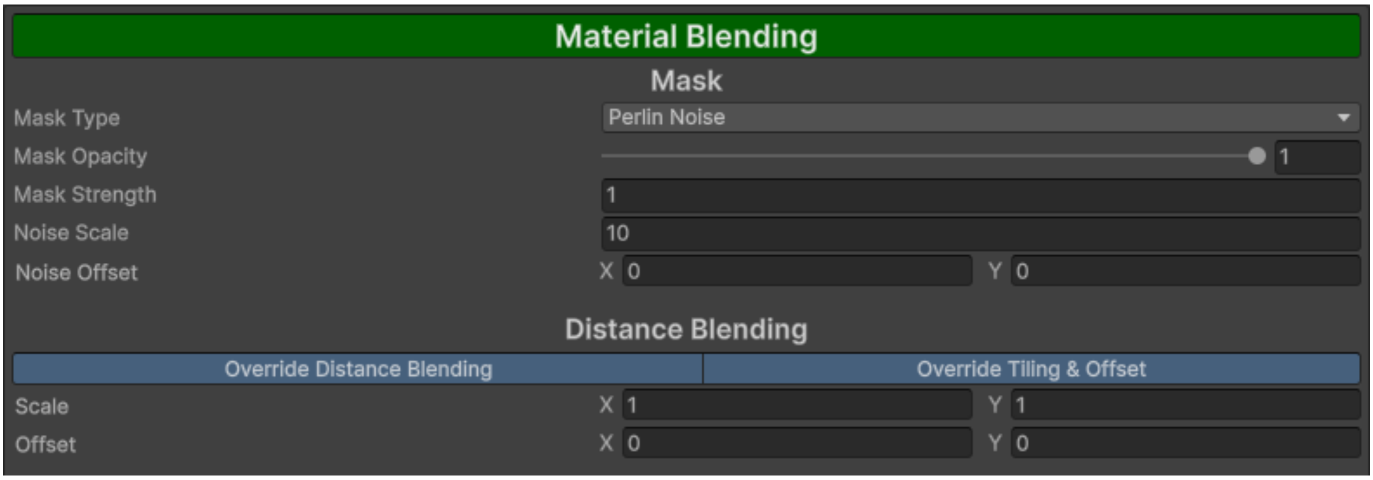


This is the material that will fade into the distance based on Distance Blend Min Max. Can be toggled by clicking the Distance Blending button

If the Blend Mode is set to Material, a separate Material Settings will be shown below for the far material

Property	Description
Blend Mode	Tiling Offset: Resamples materials with different Tiling and Offset Material: Samples a seperate far material
Distance Blend Min Max	Blend distance which the material will be sampled. Materials will be blended with regular material at min, and far material at max x: Min Distance y: Max Distance
Scale	The far tiling *Only shown if Blend mode is set to Tiling Offset*
Offset	The far offset *Only shown if Blend mode is set to Tiling Offset*

9.5 Material Blending



This is a separate material that will overlay the base and far material if set based on a mask. Can be toggled by clicking the Material Blending button

9.5.1 Mask Settings

Property	Description
Mask Type	The mask type used 0: Perlin noise 1: Simplex noise 2: Custom texture 3: Vertex Colour
Mask Opacity	Opacity of the mask and in response the blend material
Mask Strength	The higher the value, the sharper the edges and vice versa
Noise Scale	The noise tiling *Only shown if Mask Type is set to any noise*
Noise Offset	The noise offset *Only shown if Mask Type is set to any noise*
Blend Mask	Texture that is sampled as the mask for the blend material. Color from black-white represents opacity (0-1) Blend Mask (R), other channels are ignored *Only shown if Mask Type is set to Custom Texture*
Scale	The blend mask tiling *Only shown if Mask Type is set to Custom Texture*
Offset	The blend mask offset *Only shown if Mask Type is set to Custom Texture*

If mask type is set to any noise

Property	Description
Noise Scale	The noise tiling
Noise Offset	The noise offset

If mask type is set to Custom Texture

Property	Description
Blend Mask	Texture that is sampled as the mask for the blend material. Color from black-white represents opacity (0-1) Blend Mask (R), other channels are ignored
Scale	The blend mask tiling
Offset	The blend mask offset

If mask type is set to Vertex Colour

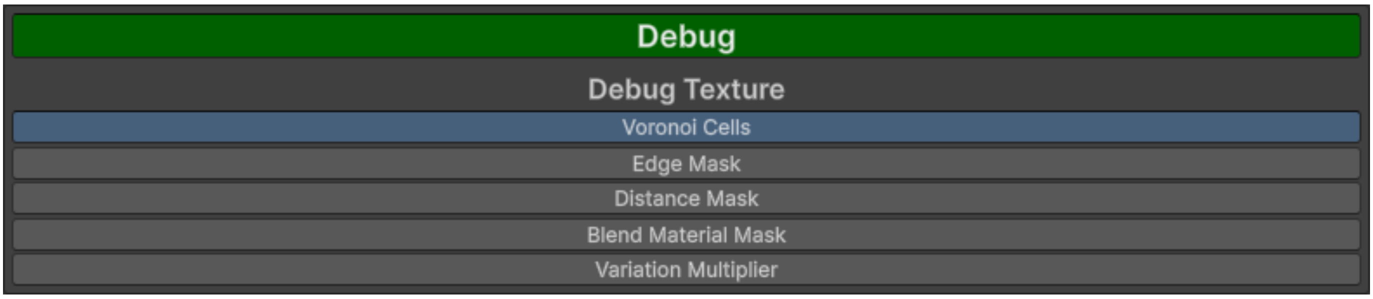
Property	Description
Mask Colour	The colour used as a mask in the vertex colour
Colour Threshold	The smoothstep min max for the colour

9.5.2 Distance Blending Settings

These are only shown if distance blending is enabled

Property	Description
Override Distance Blending	Draws the blend material on top of the far material
Override Tiling & Offset	Uses defined Tiling & Offset rather than distance blend Tiling & Offset
Scale	The blend mask distance scale *Only shown if Override Tiling & Offset is enabled*
Offset	The blend mask distance offset *Only shown if Override Tiling & Offset is enabled*

9.6 Debug

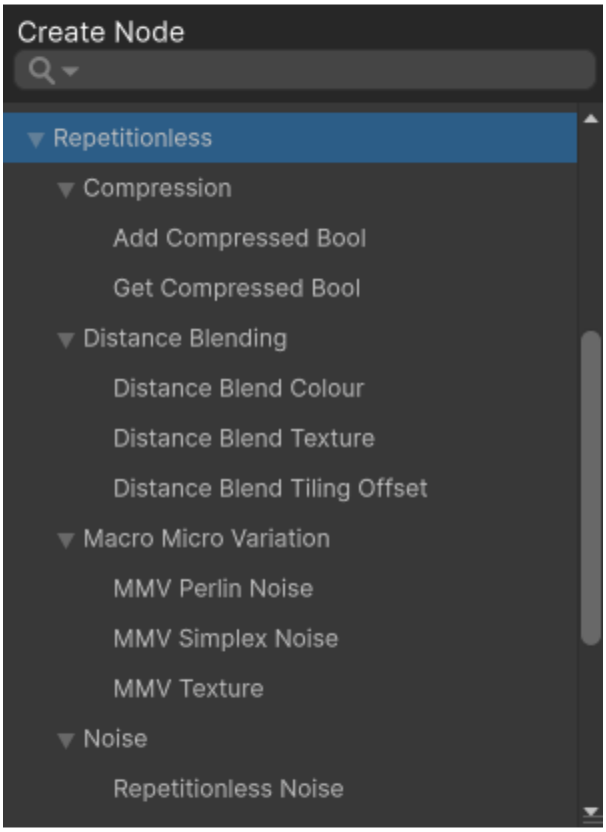


This will show the values of the selected option. Can be toggled by clicking the Debug button

Option	Description
Voronoi Cells	Outputs the value of each voronoi cell
Edge Mask	Outputs the edge mask for the voronoi cells
Distance Mask	Outputs the distance mask
Blend Material Mask	Outputs the blend material mask
Variation Multiplier	Outputs the variation multiplier

10. Shader Creation

10.1 Shader Graphs



To create a shader graph using the included sub-shaders, you can find the sub shaders in the shader graph under `Create Node > Repetitionless`

To view details about each sub-shader, view the corresponding Sub Graphs page in the contents on the left of the page

10.2 Shader Code

For creating your own shaders using the included shaders, view the respective Shader API pages in the contents on the left of the page for details on each set of functions and how to implement them into your own code

Shaders can be found at `Packages/com.williamschack.repetitionless/Shaders/HLSL`

11. Scripting

For creating your own scripts and inspectors using the included scripts, view the respective Scripting API pages in the contents on the left of the page for details on each set of functions and how to implement them into your own code

Scripts can be found at `Packages/com.williamschack.repetitionless` in the `Editor` and `Runtime` folders

To include script namespaces you must reference their respective assembly definitions in your own assembly definitions

12. Submitting Issues

If you would like to submit a Bug Report, Feature Request, or Question, you can:

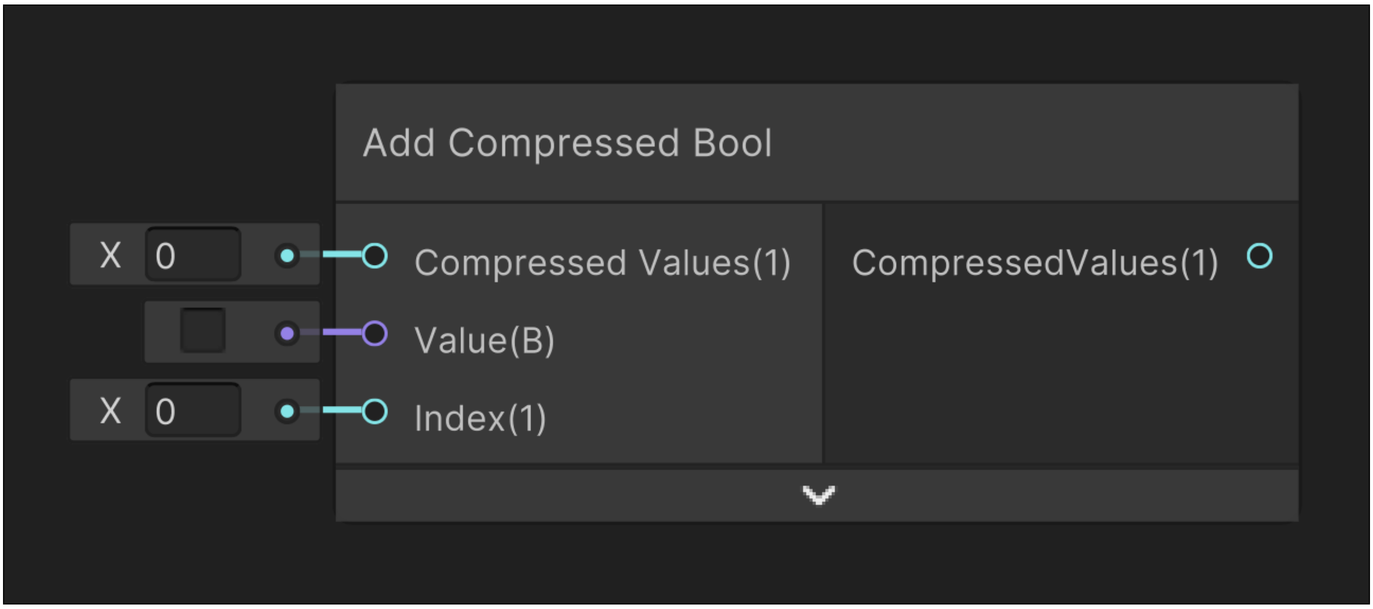
- Submit an issue to the github repo (github.com/WilliamSchack/Repetitionless-Issues)
- Email me at (support@wilschack.dev)

13. Sub Graphs

13.1 Compression

13.1.1 Add Compressed Bool

Image



Description

Overwrites a value at a given index into the input CompressedValues

Inputs

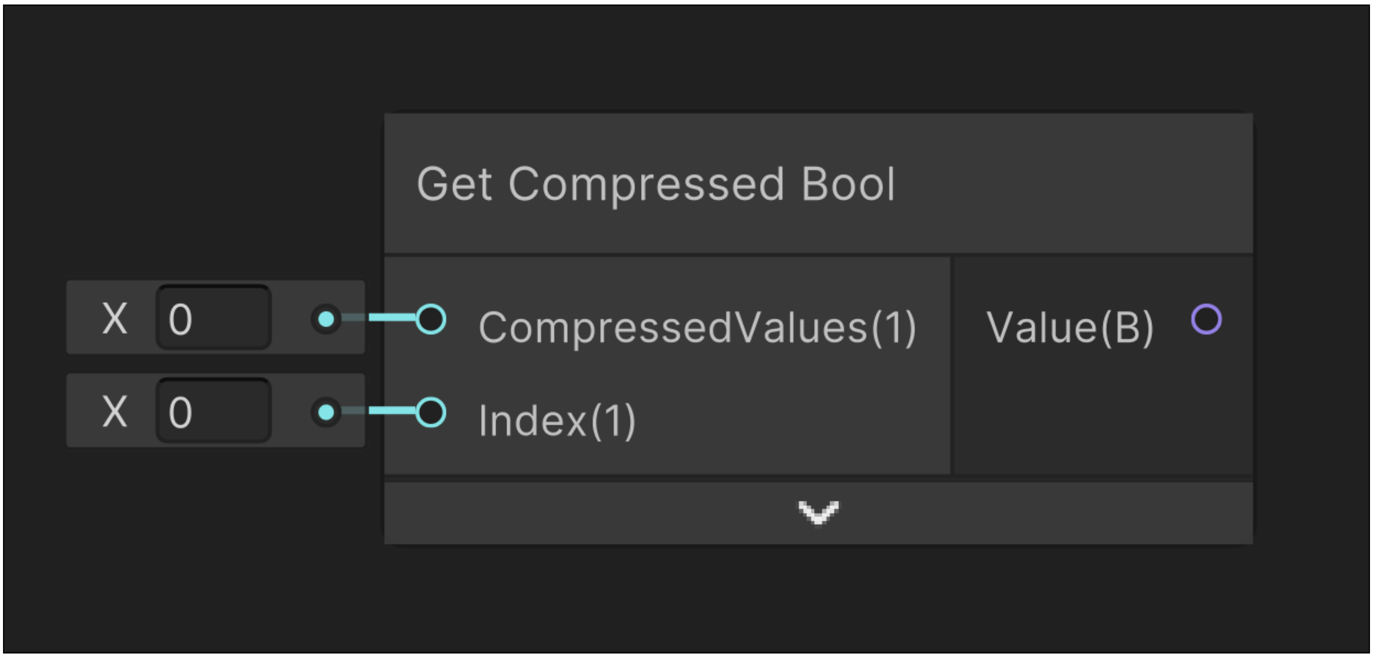
Parameter	Description
CompressedValues	Compressed int of bool values
Value	Value inserted into the CompressedValues
Index	Index that the input value will overwrite

Outputs

Output	Description
CompressedValues	New compressed integer with the added value

13.1.2 Get Compressed Bool

Image



Description

Gets a compressed value at a given index from the input CompressedValues

Inputs

Input	Description
CompressedValues	Compressed int of bool values
Index	Index of which bool value will be returned

Outputs

Output	Description
Value	Value at the given index

13.2 Distance Blending

13.2.1 Distance Blend Colour

Image

Description

Blends between two inputted colours based on the camera position

Inputs

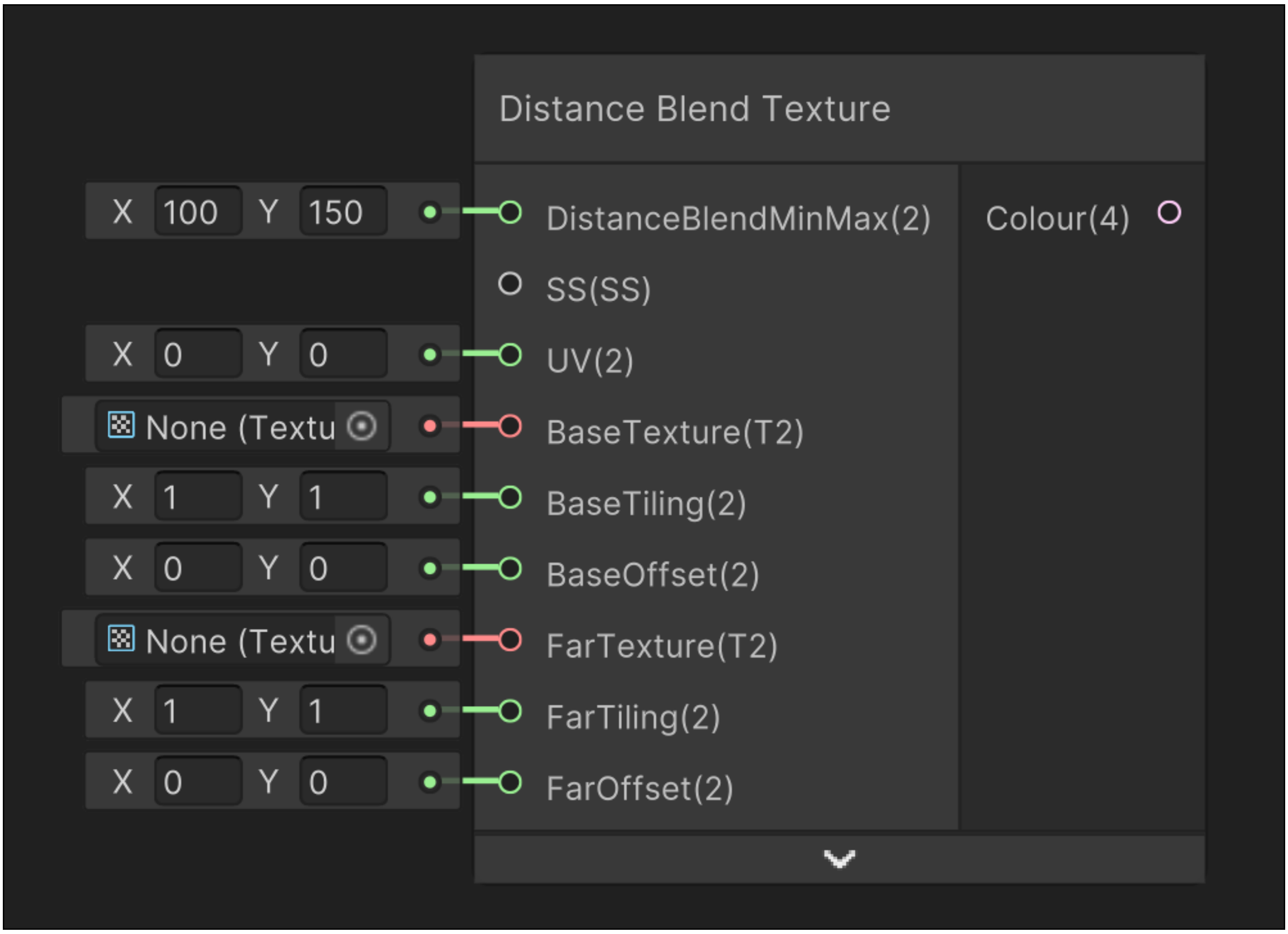
Input	Description
DistanceBlendMinMax	Blends from the BaseColour to FarColour using the world distance from the camera from X to Y
BaseColour	The colour that starts from the camera to DistanceBlendMinMax.x
FarColour	The colour that starts blending in at DistanceBlendMinMax.x, and 100% of the colour at DistanceBlendMinMax.y

Outputs

Output	Description
Colour	The output colour

13.2.2 Distance Blend Texture

Image



Description

Blends between two inputted textures based on the camera position

Inputs

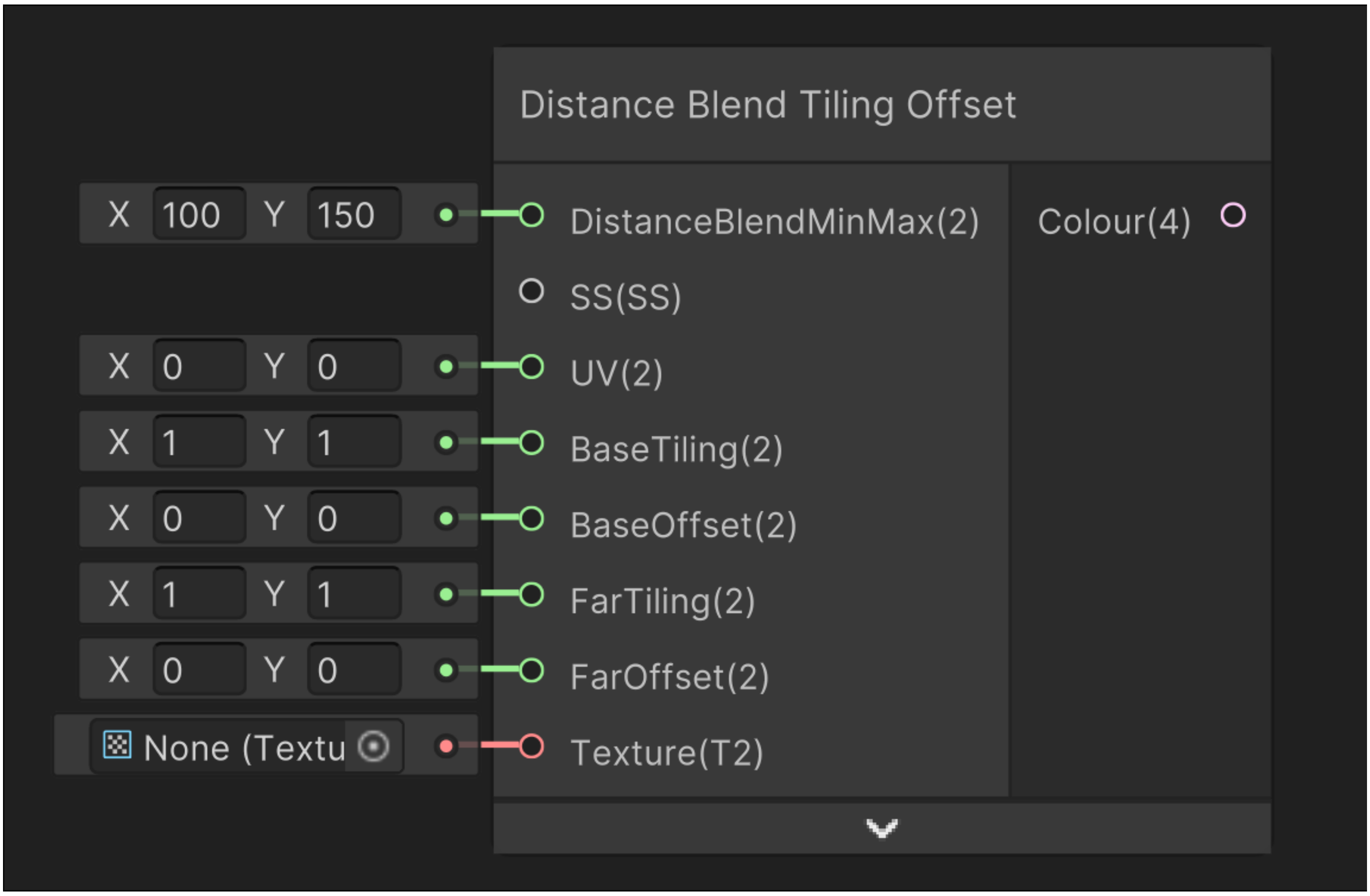
Input	Description
DistanceBlendMinMax	Blends from the BaseTexture to FarTexture using the world distance from the camera from X to Y
SS	Sampler state used for sampling textures
UV	UV used for sampling textures
BaseTexture	The texture that starts from the camera to DistanceBlendMinMax.x
BaseTiling	Tiling used when sampling the base texture
BaseOffset	Offset used when sampling the base texture
FarTexture	The texture that starts blending in at DistanceBlendMinMax.x, and 100% of the colour at DistanceBlendMinMax.y
FarTiling	Tiling used when sampling the far texture
FarOffset	Offset used when sampling the far texture

Outputs

Output	Description
Colour	The output colour

13.2.3 Distance Blend Tiling Offset

Image



Description

Blends between an inputted texture with differing tiling and offset based on the camera position

Inputs

Input	Description
DistanceBlendMinMax	Blends from the BaseTexture to FarTexture using the world distance from the camera from X to Y
SS	Sampler state used for sampling the texture
UV	UV used for sampling the texture
BaseTiling	The tiling that starts from the camera to DistanceBlendMinMax.x
BaseOffset	The offset that starts from the camera to DistanceBlendMinMax.x
FarTiling	The tiling that starts blending in at DistanceBlendMinMax.x, and 100% of the colour at DistanceBlendMinMax.y
FarOffset	The offset that starts blending in at DistanceBlendMinMax.x, and 100% of the colour at DistanceBlendMinMax.y
BaseTexture	The texture that will be sampled

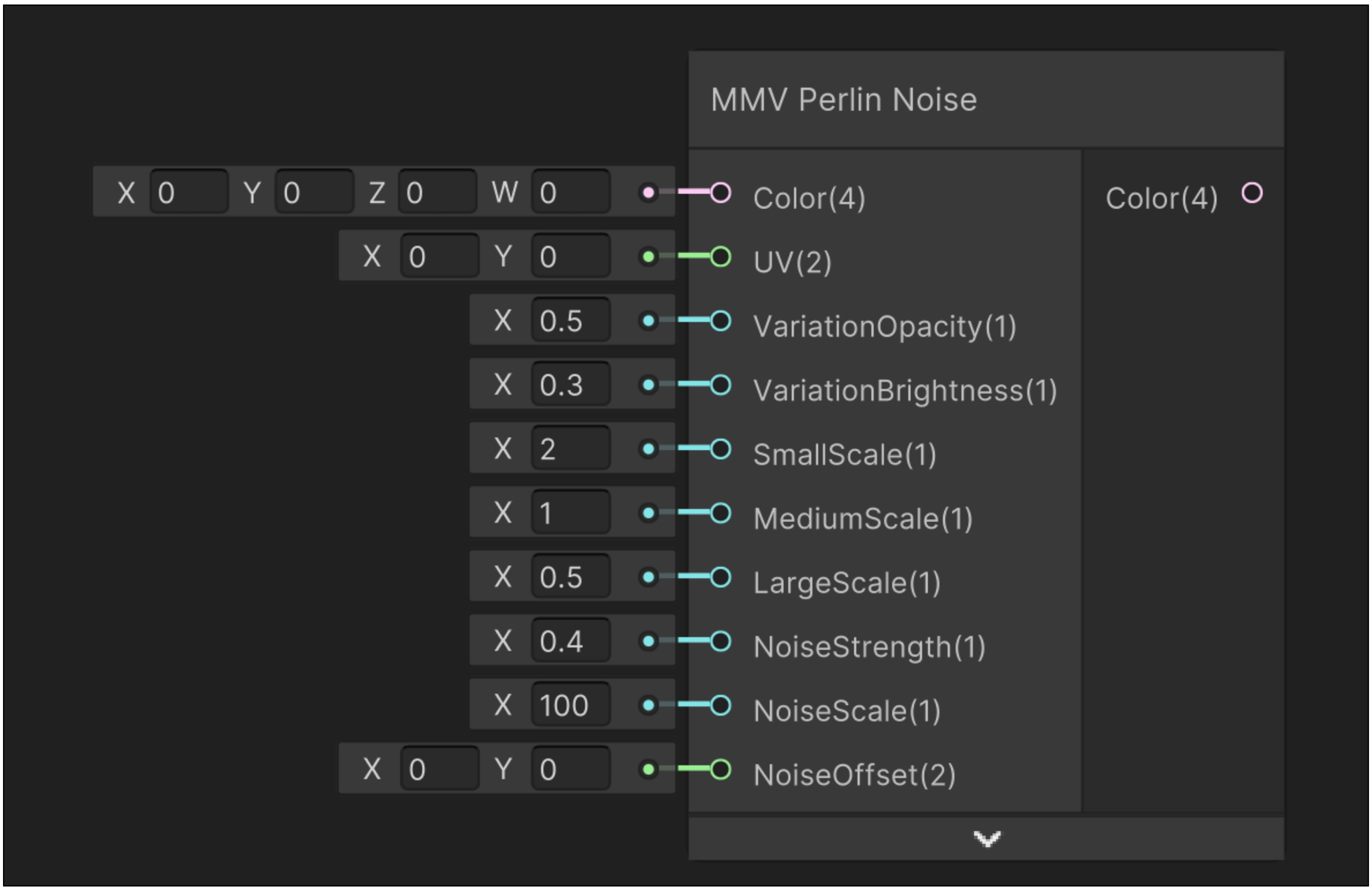
Outputs

Output	Description
Colour	The output colour

13.3 Macro Micro Variation

13.3.1 MMV Perlin Noise

Image



Description

Outputs a variation colour based on perlin noise

Inputs

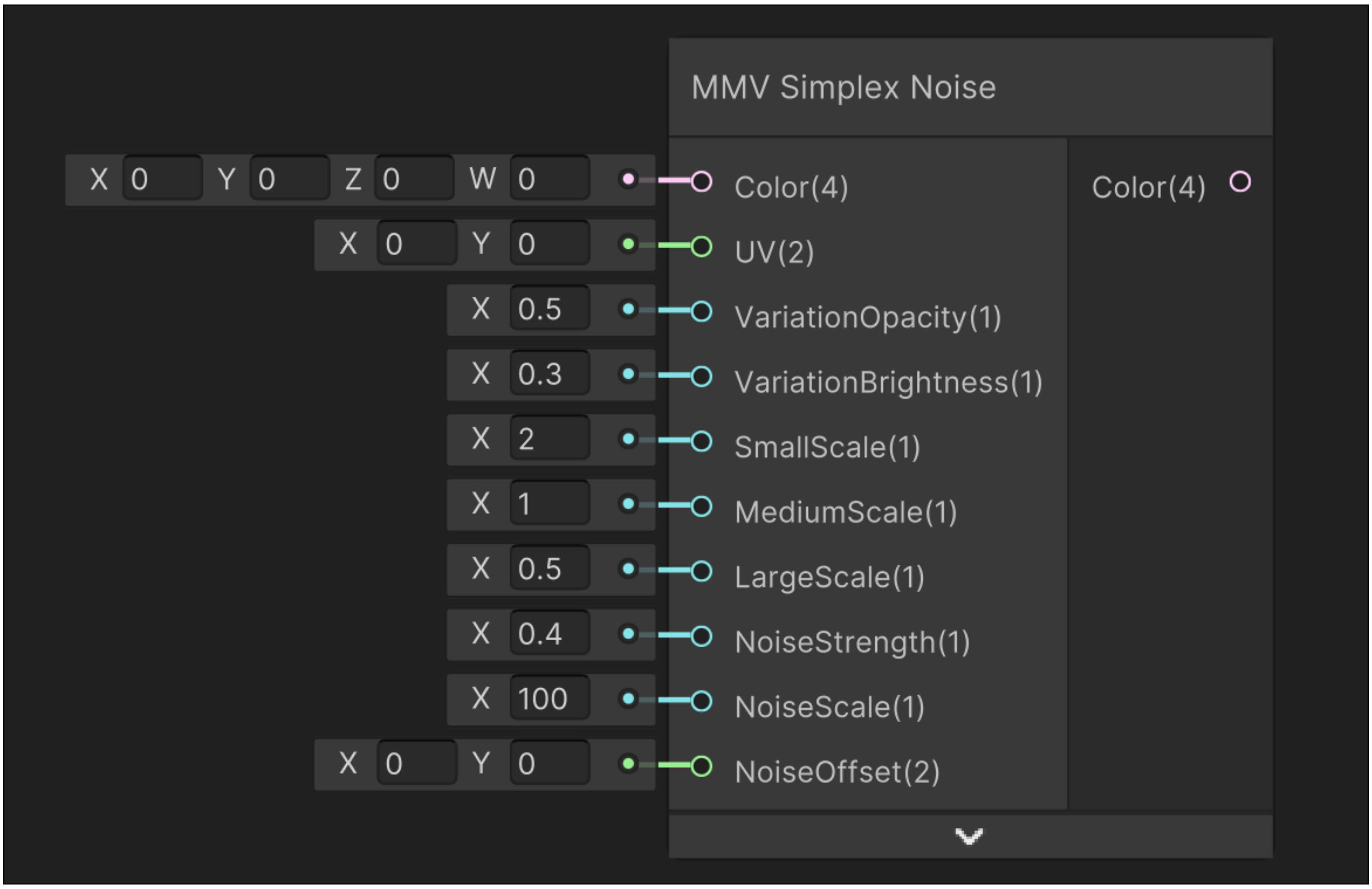
Input	Description
Color	Colour that is lerped between from this colour to variation colour
UV	UV used for sampling noise
VariationOpacity	Lerp factor from regular colour to variation colour
VariationBrightness	Brightness added onto variation colour
SmallScale	Scale for the small noise sampling
MediumScale	Scale for the medium noise sampling
LargeScale	Scale for the large noise sampling
NoiseStrength	Strength of the noise
NoiseScale	Noise Scaling
NoiseOffset	Noise Offset

Outputs

Output	Description
Colour	The output colour

13.3.2 MMV Simplex Noise

Image



Description

Outputs a variation colour based on simplex noise

Inputs

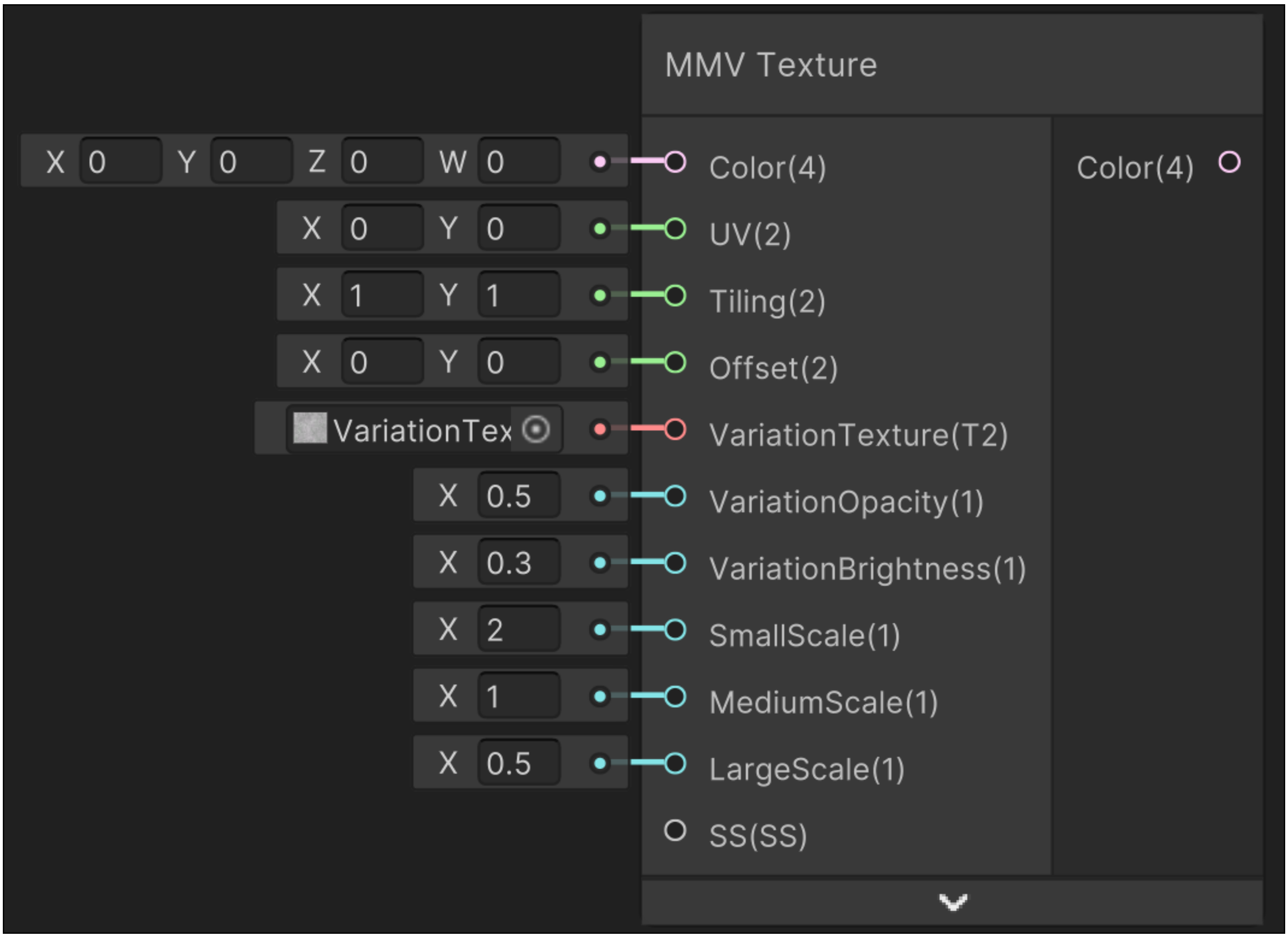
Input	Description
Color	Colour that is lerped between from this colour to variation colour
UV	UV used for sampling noise
VariationOpacity	Lerp factor from regular colour to variation colour
VariationBrightness	Brightness added onto variation colour
SmallScale	Scale for the small noise sampling
MediumScale	Scale for the medium noise sampling
LargeScale	Scale for the large noise sampling
NoiseStrength	Strength of the noise
NoiseScale	Noise Scaling
NoiseOffset	Noise Offset

Outputs

Output	Description
Colour	The output colour

13.3.3 MMV Texture

Image



Description

Outputs a variation colour based on an inputted texture

Inputs

Input	Description
Color	Colour that is lerped between from this colour to variation texture colour
UV	UV used for sampling the texture
Tiling	Texture Tiling
Offset	Texture Offset
VariationTexture	Texture to be overlayed onto the colour
VariationOpacity	Lerp factor from regular colour to variation texture
VariationBrightness	Brightness added onto variation texure
SmallScale	Scale for the small texture sampling
MediumScale	Scale for the medium texture sampling
LargeScale	Scale for the large texture sampling
SS	Sampler state used for sampling the texture

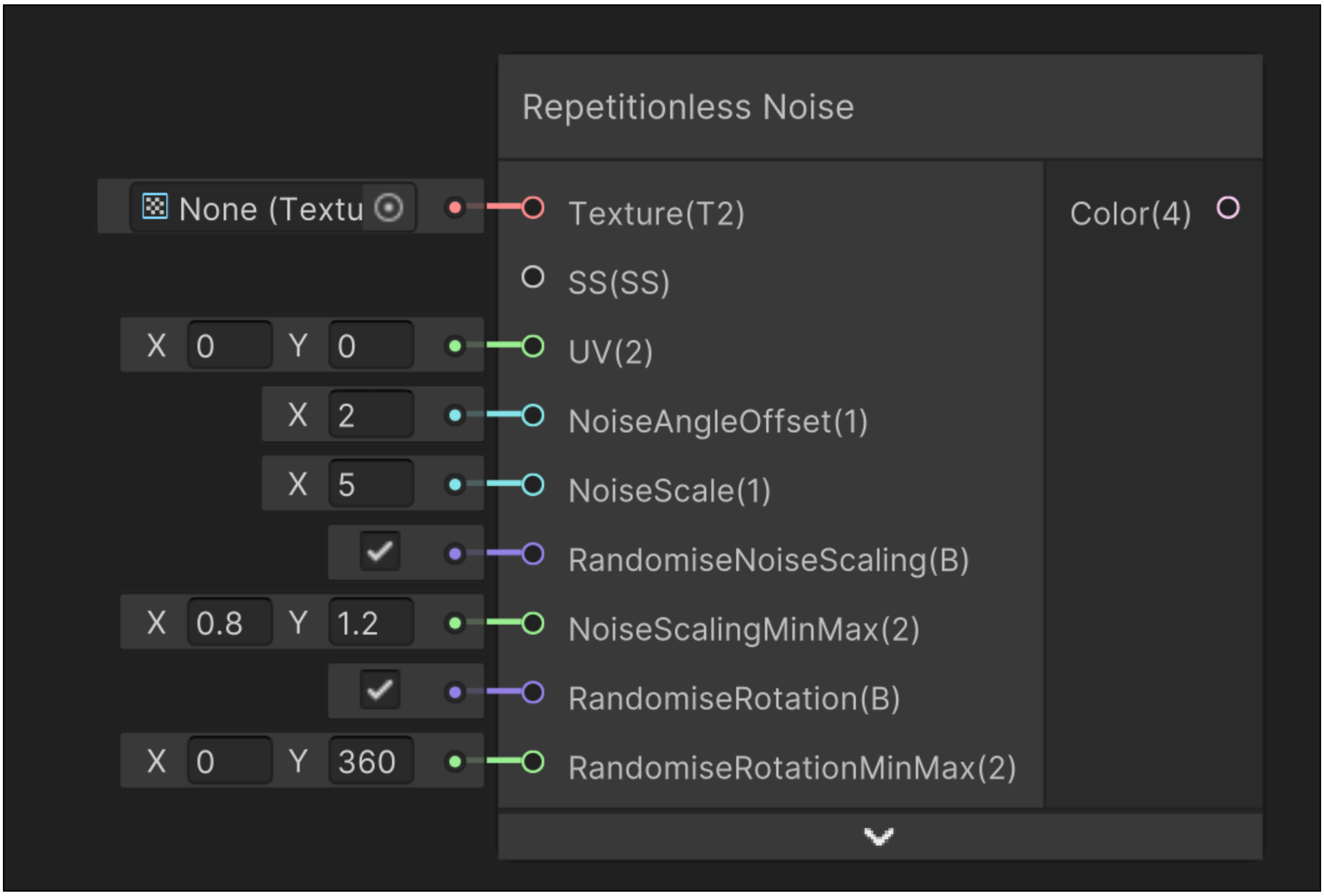
Outputs

Output	Description
Colour	The output colour

13.4 Noise

13.4.1 Repetitionless Noise

Image



Description

Samples the given texture using modified UVs based on voronoi noise

Samples the voronoi cells base and edge colour if required and lerps them together

Inputs

Input	Description
Texture	The texture to sample
SS	Sampler state used for sampling the texture
UV	UV used for sampling the texture
NoiseAngleOffset	Voronoi noise angle offset
NoiseScale	Voronoi noise scale
RandomiseNoiseScaling	If the noise scaling is randomised. NoiseScalingMinMax is ignored if disabled
NoiseScalingMinMax	The range to randomly scale the noise
RandomiseRotation	If the noise rotation is randomised. RandomiseRotationMinMax is ignored if disabled
RandomiseRotationMinMax	The range in degrees to randomly rotate the noise

Outputs

Output	Description
Colour	The output colour

14. Shader API

14.1 Main

14.1.1 Main/SampleRepetitionlessMaterial

Description

Used to sample a repetitionless material

GetRepetitionlessMaterialColor()

DECLARATION

```
void GetRepetitionlessMaterialColor(
    SamplerState SS, float2 UV, float3 WorldNormalVector,
    int SurfaceType, int DebuggingIndex,

    int ArrayLayerIndex,
    Texture2DArray AVTextures,
    Texture2DArray NSOTextures,
    Texture2DArray EMTextures,
    int AssignedAVTextures[3],
    int AssignedNSOTextures[3],
    int AssignedEMTextures[3],

    in RepetitionlessMaterialData MaterialData,

    out float4 AlbedoColorOut,
    out float3 NormalVectorOut,
    out float MetallicOut,
    out float SmoothnessOut,
    out float OcclussionOut,
    out float3 EmissionColorOut
)
```

PARAMETERS

Parameter	Description
SS	Sampler state used for sampling textures
UV	The UV used for sampling textures and noise
WorldNormalVector	The current world normal vector
SurfaceType	The surface type of the material as shown in the inspector 0: Opaque 1: Cutout 2: Transparent
DebuggingIndex	The selected debugging type as shown in the inspector *Anything outside the below range is disabled* 0: Voronoi Cells 1: Edge Mask 2: Distance Mask 3: Blend Material Mask 4: Variation Multiplier
LayerIndex	The constant index of the layer in the texture arrays
AVTextures	The texture array storing the AVTextures
NSOTextures	The texture array storing the NSOTextures
EMTextures	The texture array storing the EMTextures
AssignedAVTextures	The assigned AVTextures, each value in the array is 32 bools
AssignedNSOTextures	The assigned NSOTextures, each value in the array is 32 bools
AssignedEMTextures	The assigned EMTextures, each value in the array is 32 bools
MaterialData	The material data that will be used

RETURNS

Output	Description
AlbedoColorOut	The output albedo colour
NormalVectorOut	The output normal vector
MetallicOut	The output metallic colour
SmoothnessOut	The output smoothness colour
OcclussionOut	The output occlussion colour
EmissionColorOut	The output emission colour

DESCRIPTION

Samples a RepetitionlessMaterial

14.1.2 Main/SampleRepetitionlessLayer

Description

Used to sample a repetitionless layer

SampleRepetitionlessLayer()

DECLARATION

```
void SampleRepetitionlessLayer_float(
    SamplerState SS, float2 UV, float3 WorldNormalVector,
    float3 WorldPosition, float3 CameraPosition,
    int SurfaceType, int DebuggingIndex,

    int LayerIndex,
    Texture2D PropertiesTexture,
    Texture2D AssignedTexturesTexture,

    Texture2DArray AVTextures,
    Texture2DArray NSOTextures,
    Texture2DArray EMTextures,
    Texture2DArray BMTextures,

    out float4 AlbedoColorOut,
    out float3 NormalVectorOut,
    out float MetallicOut,
    out float SmoothnessOut,
    out float OcclussionOut,
    out float3 EmissionColorOut
)
```

PARAMETERS

Parameter	Description
SS	Sampler state used for sampling textures
UV	The UV used for sampling textures and noise
WorldNormalVector	The current world normal vector
WorldPosition	The current world position
CameraPosition	The current camera position
SurfaceType	The surface type of the material as shown in the inspector 0: Opaque 1: Cutout 2: Transparent
DebuggingIndex	The selected debugging type as shown in the inspector *Anything outside the below range is disabled* 0: Voronoi Cells 1: Edge Mask 2: Distance Mask 3: Blend Material Mask 4: Variation Multiplier
LayerIndex	The index of the layer being sampled
PropertiesTexture	The texture storing the material properties
AssignedTexturesTexture	The texture storing all the arrays assigned textures
AVTextures	The texture array storing the AVTextures
NSOTextures	The texture array storing the NSOTextures
EMTextures	The texture array storing the EMTextures
BMTextures	The texture array storing the BMTextures

RETURNS

Output	Description
AlbedoColorOut	The output albedo colour
NormalVectorOut	The output normal vector
MetallicOut	The output metallic colour
SmoothnessOut	The output smoothness colour
OcclussionOut	The output occlussion colour
EmissionColorOut	The output emission colour

DESCRIPTION

Samples a RepetitionlessLayer, automatically blending between and handling each material in the layer

SampleRepetitionlessLayer()

DECLARATION

```
void SampleRepetitionlessLayer_float(
    SamplerState SS, float2 UV, float3 WorldNormalVector,
    float3 WorldPosition, float3 CameraPosition,
    int SurfaceType, int DebuggingIndex,

    int LayerIndex,
    Texture2D PropertiesTexture,
    int AssignedAVTextures0,
    int AssignedAVTextures1,
    int AssignedAVTextures2,
    int AssignedNSOTextures0,
    int AssignedNSOTextures1,
    int AssignedNSOTextures2,
    int AssignedEMTextures0,
    int AssignedEMTextures1,
    int AssignedEMTextures2,
    int AssignedBMTextures0,

    Texture2DArray AVTextures,
    Texture2DArray NSOTextures,
    Texture2DArray EMTextures,
    Texture2DArray BMTextures,

    out float4 AlbedoColorOut,
    out float3 NormalVectorOut,
    out float MetallicOut,
    out float SmoothnessOut,
    out float OcclussionOut,
    out float3 EmissionColorOut
)
```

PARAMETERS

Parameter	Description
SS	Sampler state used for sampling textures
UV	The UV used for sampling textures and noise
WorldNormalVector	The current world normal vector
WorldPosition	The current world position
CameraPosition	The current camera position
SurfaceType	The surface type of the material as shown in the inspector 0: Opaque 1: Cutout 2: Transparent
DebuggingIndex	The selected debugging type as shown in the inspector *Anything outside the below range is disabled* 0: Voronoi Cells 1: Edge Mask 2: Distance Mask 3: Blend Material Mask 4: Variation Multiplier
LayerIndex	The index of the layer being sampled
PropertiesTexture	The texture storing the material properties
AssignedAVTextures0	The assigned AVTextures (0-31)
AssignedAVTextures1	The assigned AVTextures (32-63)
AssignedAVTextures2	The assigned AVTextures (64-95)
AssignedNSOTextures0	The assigned NSOTextures (0-31)
AssignedNSOTextures1	The assigned NSOTextures (32-63)
AssignedNSOTextures2	The assigned NSOTextures (64-95)
AssignedEMTextures0	The assigned EMTextures (0-31)
AssignedEMTextures1	The assigned EMTextures (32-63)
AssignedEMTextures2	The assigned EMTextures (64-95)
AssignedBMTextures0	The assigned BMTextures (0-31)
AVTextures	The texture array storing the AVTextures
NSOTextures	The texture array storing the NSOTextures
EMTextures	The texture array storing the EMTextures
BMTextures	The texture array storing the BMTextures

RETURNS

Output	Description
AlbedoColorOut	The output albedo colour
NormalVectorOut	The output normal vector
MetallicOut	The output metallic colour
SmoothnessOut	The output smoothness colour
OcclussionOut	The output occlussion colour
EmissionColorOut	The output emission colour

DESCRIPTION

Samples a RepetitionlessLayer, automatically blending between and handling each material in the layer

14.1.3 Main/SampleRepetitionlessTerrain

Description

Used to sample a repetitionless terrain

SampleRepetitionlessLayer()

DECLARATION

```
void SampleRepetitionlessLayer_float(
    SamplerState SS, float2 UV, float3 WorldNormalVector,
    float3 WorldPosition, float3 CameraPosition,
    int SurfaceType, int DebuggingIndex,

    int LayersCount,
    Texture2D PropertiesTexture,
    Texture2D AssignedTexturesTexture,

    Texture2DArray AVTextures,
    Texture2DArray NSOTextures,
    Texture2DArray EMTextures,
    Texture2DArray BMTextures,

    out float4 AlbedoColorOut,
    out float3 NormalVectorOut,
    out float MetallicOut,
    out float SmoothnessOut,
    out float OcclussionOut,
    out float3 EmissionColorOut
)
```

PARAMETERS

Parameter	Description
SS	Sampler state used for sampling textures
UV	The UV used for sampling textures and noise
WorldNormalVector	The current world normal vector
WorldPosition	The current world position
CameraPosition	The current camera position
SurfaceType	The surface type of the material as shown in the inspector 0: Opaque 1: Cutout 2: Transparent
DebuggingIndex	The selected debugging type as shown in the inspector *Anything outside the below range is disabled* 0: Voronoi Cells 1: Edge Mask 2: Distance Mask 3: Blend Material Mask 4: Variation Multiplier
LayersCount	The amount of layers that will be sampled
PropertiesTexture	The texture storing the material properties
AssignedTexturesTexture	The texture storing all the arrays assigned textures
AVTextures	The texture array storing the AVTextures
NSOTextures	The texture array storing the NSOTextures
EMTextures	The texture array storing the EMTextures
BMTextures	The texture array storing the BMTextures

RETURNS

Output	Description
AlbedoColorOut	The output albedo colour
NormalVectorOut	The output normal vector
MetallicOut	The output metallic colour
SmoothnessOut	The output smoothness colour
OcclussionOut	The output occlussion colour
EmissionColorOut	The output emission colour

DESCRIPTION

Samples a multiple RepetitionlessLayers using the control textures from the terrain and clipping using the holes texture from the terrain.

14.2 Noise

14.2.1 Noise/VoronoiNoise2D

Description

Based on code by Inigo Quilez

`https://iquilezles.org/articles/voronoilines/`

Used for voronoi noise

VoronoiNoise()

DECLARATION

```
void VoronoiNoise(float2 UV, float AngleOffset, float CellDensity, out float DistFromCenter, out float DistFromEdge, out float Cells)
```

PARAMETERS

Parameter	Description
UV	Coordinate to sample the noise
AngleOffset	Varies rotation of the noise
CellDensity	Changes amount of cells

RETURNS

Output	Description
DistFromCenter	The distance to the closest cell
DistFromEdge	The distance to the closest edge
Cells	The current noise value

DESCRIPTION

Voronoi noise

14.2.2 Keijiro

Noise/Keijiro/ClassicNoise2D

DESCRIPTION

Taken from keijiro's NoiseShader repo
`https://github.com/keijiro/NoiseShader`

Used for perlin noise

CLASSICNOISE()

Declaration

```
float ClassicNoise(float2 p)
```

Parameters

Parameter	Description
p	Coordinate to sample the noise

Returns

The noise value

Description

Classic Perlin noise

PERIODICNOISE()

Declaration

```
float PeriodicNoise(float2 p, float2 rep)
```

Parameters

Parameter	Description
p	Coordinate to sample the noise
rep	Repeat interval

Returns

The noise value

Description

Classic Perlin noise, periodic variant

Noise/Keijiro/SimplexNoise2D

DESCRIPTION

Taken from keijiro's NoiseShader repo

`https://github.com/keijiro/NoiseShader`

Used for simplex noise

SIMPLEXNOISE()

Declaration

```
float SimplexNoise(float2 v)
```

Parameters

Parameter	Description
v	Coordinate to sample the noise

Returns

The noise value

Description

Simplex noise

14.3 RepetitionlessHelpers

14.3.1 RepetitionlessHelpers/DistanceBlend

Description

Helper for changing and blending values between and past a distance range

Used for sampling different textures or colours based on distance to the camera

CalculateFarDistance()

DECLARATION

```
float CalculateFarDistance(float3 WorldPosition, float3 CameraPosition, float2 DistanceBlendMinMax)
```

PARAMETERS

Parameter	Description
WorldPosition	The current world position
CameraPosition	The current camera position
DistanceBlendMinMax	Lerps from the 0 to 1 using the world distance from the camera from X to Y

RETURNS

The normalised distance between the inputted min max based on the camera position

DESCRIPTION

Calculates a normalised distance between the inputted min max based on the camera position

DistanceBlendTilingOffset_float()

DECLARATION

```
void DistanceBlendTilingOffset_float(
    float3 WorldPosition, float3 CameraPosition,
    float2 DistanceBlendMinMax,

    SamplerState SS, float2 UV,
    float4 BaseTilingOffset, float4 FarTilingOffset,
    Texture2D Texture,

    out float4 ColourOut
)
```

PARAMETERS

Parameter	Description
WorldPosition	The current world position
CameraPosition	The current camera position
DistanceBlendMinMax	Lerps from the 0 to 1 using the world distance from the camera from X to Y
SS	Sampler state used for sampling textures
UV	UV used for sampling textures
BaseTilingOffset	The tiling and offset that starts from the camera to DistanceBlendMinMax.x
FarTilingOffset	The tiling and offset that starts blending in at DistanceBlendMinMax.x, and 100% of the colour at DistanceBlendMinMax.y
Texture	The texture that will be sampled

RETURNS

Output	Description
ColourOut	The output colour

DESCRIPTION

Blends between a texture with differing tiling offset based on the camera position

Used in the Distance Blend Tiling Offset sub graph

DistanceBlendColour_float(()

DECLARATION

```
void DistanceBlendColour_float(
    float3 WorldPosition, float3 CameraPosition,
    float2 DistanceBlendMinMax,

    float4 BaseColour, float4 FarColour,

    out float4 ColourOut
)
```

PARAMETERS

Parameter	Description
WorldPosition	The current world position
CameraPosition	The current camera position
DistanceBlendMinMax	Lerps from the 0 to 1 using the world distance from the camera from X to Y
BaseTilingOffset	The colour that starts from the camera to DistanceBlendMinMax.x
FarTilingOffset	The colour that starts blending in at DistanceBlendMinMax.x, and 100% of the colour at DistanceBlendMinMax.y

RETURNS

Output	Description
ColourOut	The output colour

DESCRIPTION

Blends between two inputted colours based on the camera position

Used in the Distance Blend Colour, Distance Blend Texture sub graphs

14.3.2 RepetitionlessHelpers/MacroMicroVariation

Description

Gets variation for a material based on noise or a custom texture

MacroMicroVariationTexture()

DECLARATION

```
float MacroMicroVariationTexture(
    float SmallScale,
    float MediumScale,
    float LargeScale,

    float VariationBrightness,
    Texture2D Texture,
    SamplerState SS,

    float2 UV,
    float2 Tiling = float2(1, 1),
    float2 Offset = float2(0, 0)
)
```

PARAMETERS

Parameter	Description
SmallScale	Scale for the small texture sampling
MediumScale	Scale for the medium texture sampling
LargeScale	Scale for the large texture sampling
VariationBrightness	Brightness added onto variation texure
Texture	Texture to be sampled
SS	Sampler state used for sampling the texture
UV	UV used for sampling the texture
Tiling	Texture Tiling
Offset	Texture Offset

RETURNS

The variation multiplier based on the texture samples

DESCRIPTION

Samples a given texture and turns it into a variation multiplier

MacroMicroVariationPerlinNoise()

DECLARATION

```
float MacroMicroVariationPerlinNoise(
    float SmallScale,
    float MediumScale,
    float LargeScale,

    float VariationBrightness,

    float NoiseStrength,
    float2 UV,
    float NoiseScale = 1,
```

```
    float2 NoiseOffset = float2(0, 0)
)
```

PARAMETERS

Parameter	Description
SmallScale	Scale for the small texture sampling
MediumScale	Scale for the medium texture sampling
LargeScale	Scale for the large texture sampling
VariationBrightness	Brightness added onto variation texure
Texture	Texture to be sampled
NoiseStrength	Strength of the noise
UV	UV used for sampling the noise
NoiseScale	Noise Scaling
NoiseOffset	Noise Offset

RETURNS

The variation multiplier based on the noise samples

DESCRIPTION

Samples perlin noise and turns it into a variation multiplier

---

MacroMicroVariationSimplexNoise()

DECLARATION

```
float MacroMicroVariationSimplexNoise(
    float SmallScale,
    float MediumScale,
    float LargeScale,

    float VariationBrightness,

    float NoiseStrength,
    float2 UV,
    float NoiseScale = 1,
    float2 NoiseOffset = float2(0, 0)
)
```

PARAMETERS

Parameter	Description
SmallScale	Scale for the small texture sampling
MediumScale	Scale for the medium texture sampling
LargeScale	Scale for the large texture sampling
VariationBrightness	Brightness added onto variation texure
Texture	Texture to be sampled
NoiseStrength	Strength of the noise
UV	UV used for sampling the noise
NoiseScale	Noise Scaling
NoiseOffset	Noise Offset

RETURNS

The variation multiplier based on the noise samples

DESCRIPTION

Samples simplex noise and turns it into a variation multiplier

---

MacroMicroVariationTexture_float()

DECLARATION

```
void MacroMicroVariationTexture_float(
    float4 InputColor,

    float SmallScale,
    float MediumScale,
    float LargeScale,

    float VariationBrightness,
    float VariationOpacity,
    Texture2D VariationTexture,

    SamplerState SS,
    float2 UV,
    float2 Tiling,
    float2 Offset,

    out float4 OutputColor
)
```

PARAMETERS

Parameter	Description
InputColor	Colour that is lerped between from this colour to variation texture colour
SmallScale	Scale for the small texture sampling
MediumScale	Scale for the medium texture sampling
LargeScale	Scale for the large texture sampling
VariationBrightness	Brightness added onto variation texure
VariationOpacity	Lerp factor from regular colour to variation texture
VariationTexture	Texture to be overlayed onto the colour
SS	Sampler state used for sampling the texture
UV	UV used for sampling the texture
Tiling	Texture Tiling
Offset	Texture Offset

RETURNS

Output	Description
OutputColor	The variation multiplier based on the texture samples

DESCRIPTION

Outputs a variation colour based on an inputted texture

Used in the MMV Texture sub graph

MacroMicroVariationPerlinNoise_float()

DECLARATION

```
void MacroMicroVariationPerlinNoise_float(
    float4 InputColor,

    float SmallScale,
    float MediumScale,
    float LargeScale,

    float VariationBrightness,
    float VariationOpacity,

    float NoiseStrength,
    float NoiseScale,
    float2 NoiseOffset,
    float2 UV,

    out float4 OutputColor
)
```

PARAMETERS

Input	Description
InputColor	Colour that is lerped between from this colour to variation colour
SmallScale	Scale for the small noise sampling
MediumScale	Scale for the medium noise sampling
LargeScale	Scale for the large noise sampling
VariationBrightness	Brightness added onto variation colour
VariationOpacity	Lerp factor from regular colour to variation colour
NoiseStrength	Strength of the noise
NoiseScale	Noise Scaling
NoiseOffset	Noise Offset
UV	UV used for sampling noise

RETURNS

Output	Description
OutputColor	The variation multiplier based on the texture samples

DESCRIPTION

Outputs a variation colour based on perlin noise

Used in the MMV Perlin Noise sub graph

MacroMicroVariationSimplexNoise_float()

DECLARATION

```
void MacroMicroVariationSimplexNoise_float(
    float4 InputColor,

    float SmallScale,
    float MediumScale,
    float LargeScale,

    float VariationBrightness,
    float VariationOpacity,

    float NoiseStrength,
    float NoiseScale,
    float2 NoiseOffset,
    float2 UV,

    out float4 OutputColor
)
```

PARAMETERS

Input	Description
InputColor	Colour that is lerped between from this colour to variation colour
SmallScale	Scale for the small noise sampling
MediumScale	Scale for the medium noise sampling
LargeScale	Scale for the large noise sampling
VariationBrightness	Brightness added onto variation colour
VariationOpacity	Lerp factor from regular colour to variation colour
NoiseStrength	Strength of the noise
NoiseScale	Noise Scaling
NoiseOffset	Noise Offset
UV	UV used for sampling noise

RETURNS

Output	Description
OutputColor	The variation multiplier based on the texture samples

DESCRIPTION

Samples simplex noise and turns it into a variation multiplier

Used in the MMV Simplex Noise sub graph

14.3.3 RepetitionlessHelpers/RepetitionlessNoise

Description

Takes and modifies UVs, returning those and noise masks and data that is used to remove repetition from materials

---

GetRepetitionlessNoiseUVs()

DECLARATION

```
void GetRepetitionlessNoiseUVs(
    float2 UV,

    float NoiseAngleOffset,
    float NoiseScale,
    bool RandomiseNoiseScaling,
    float2 NoiseScalingMinMax,

    bool RandomiseRotation,
    float2 RandomiseRotationMinMax,

    out float VoronoiCells,
    out float EdgeMask,
    out float2 EdgeUV,
    out float2 TransformedUV
)
```

PARAMETERS

Parameter	Description
UV	UV used for sampling the noise
NoiseAngleOffset	Voronoi noise angle offset
NoiseScale	Voronoi noise scale
RandomiseNoiseScaling	If the noise scaling is randomised. NoiseScalingMinMax is ignored if disabled
NoiseScalingMinMax	The range to randomly scale the noise
RandomiseRotation	If the noise rotation is randomised. RandomiseRotationMinMax is ignored if disabled
RandomiseRotationMinMax	The range in degrees to randomly rotate the noise

RETURNS

Output	Description
VoronoiCells	The current voronoi cell noise value
EdgeMask	The edge mask of each cell
EdgeUV	The UV to sample edges
TransformedUV	The UV to sample cells

DESCRIPTION

Gets UVs based on voronoi noise

GetRepetitionlessNoiseUVs()

DECLARATION

```
void GetRepetitionlessNoiseUVs(
    float2 UV,
```

```
    float NoiseScale,
    bool RandomiseNoiseScaling,
    float2 NoiseScalingMinMax,

    bool RandomiseRotation,
    float2 RandomiseRotationMinMax,

    Texture2D NoiseTexture,
    int TextureResolution,

    out float VoronoiCells,
    out float EdgeMask,
    out float2 EdgeUV,
    out float2 TransformedUV
)
```

PARAMETERS

Parameter	Description
UV	UV used for sampling the noise
NoiseScale	Voronoi noise scale
RandomiseNoiseScaling	If the noise scaling is randomised. NoiseScalingMinMax is ignored if disabled
NoiseScalingMinMax	The range to randomly scale the noise
RandomiseRotation	If the noise rotation is randomised. RandomiseRotationMinMax is ignored if disabled
RandomiseRotationMinMax	The range in degrees to randomly rotate the noise
NoiseTexture	The noise texture to load from
TextureResolution	The resolution of the texture. Should equal to the width and height

RETURNS

Output	Description
VoronoiCells	The current voronoi cell noise value
EdgeMask	The edge mask of each cell
EdgeUV	The UV to sample edges
TransformedUV	The UV to sample cells

DESCRIPTION

Gets UVs based on a voronoi texture

AddRepetitionlessNoise_float()

DECLARATION

```
void AddRepetitionlessNoise_float(
    Texture2D InputTexture,
    SamplerState SS,
    float2 UV,

    float NoiseAngleOffset,
    float NoiseScale,
    bool RandomiseNoiseScaling,
    float2 NoiseScalingMinMax,

    bool RandomiseRotation,
    float2 RandomiseRotationMinMax,

    out float4 OutputColor
)
```

PARAMETERS

Parameter	Description
Texture	The texture to sample
SS	Sampler state used for sampling the texture
UV	UV used for sampling the texture
NoiseAngleOffset	Voronoi noise angle offset
NoiseScale	Voronoi noise scale
RandomiseNoiseScaling	If the noise scaling is randomised. NoiseScalingMinMax is ignored if disabled
NoiseScalingMinMax	The range to randomly scale the noise
RandomiseRotation	If the noise rotation is randomised. RandomiseRotationMinMax is ignored if disabled
RandomiseRotationMinMax	The range in degrees to randomly rotate the noise

RETURNS

Output	Description
OutputColor	The output colour

DESCRIPTION

Samples the given texture using modified UVs based on voronoi noise.

Samples the voronoi cells base and edge colour if required and lerps them together

Used in the Repetitionless Noise sub graph

14.3.4 RepetitionlessHelpers/RepetitionlessTextureUtilities

Description

Samples textures at the center and edges if required of modified UVs based on voronoi noise. UVs are expected to come from RepetitionlessNoise

---

SampleRepetitionlessTexture()

DECLARATION

```
float4 SampleRepetitionlessTexture(
    Texture2D Texture,
    SamplerState SS,

    float EdgeMask,
    float2 EdgeUV,
    float2 TransformedUV,
    bool SampleEdge,

    bool NormalMap = false,
    float NormalStrength = 1.0
)
```

PARAMETERS

Parameter	Description
Texture	The texture to sample
SS	Sampler state used for sampling the texture
EdgeUV	The UV to sample edges
TransformedUV	The UV to sample cells
SampleEdge	If the edge is being sampled
NormalMap	If this texture is a normal map
NormalStrength	The normal strength *Only used if NormalMap is enabled*

RETURNS

The output colour

DESCRIPTION

Samples the base and edge colour if required and lerps them together using a regular texture

---

SampleRepetitionlessArrayTexture()

DECLARATION

```
float4 SampleRepetitionlessArrayTexture(
    Texture2DArray TextureArray,
    int AssignedTextures[3],
    int ConstantIndex,
    SamplerState SS,

    float EdgeMask,
    float2 EdgeUV,
    float2 TransformedUV,
    bool SampleEdge
)
```

PARAMETERS

Parameter	Description
TextureArray	The texture to sample
AssignedTextures	The compressed assigned textures that contains which textures are assigned in the array, each value in the array is 32 bools
ConstantIndex	The constant index of the texture in the array *(Doesnt care about how many textures are in the array, think of it as a constant within the max texture count)*
SS	Sampler state used for sampling the texture
EdgeUV	The UV to sample edges
TransformedUV	The UV to sample cells
SampleEdge	If the edge is being sampled
NormalMap	If this texture is a normal map
NormalStrength	The normal strength *Only used if NormalMap is enabled*

RETURNS

The output colour

DESCRIPTION

Samples the base and edge colour if required and lerps them together using a texture array

Normal maps do not work properly in an array so they are not allowed

14.3.5 RepetitionlessHelpers/GetArrayAssignedTextures

Description

Loads the array assigned textures from a texture

GetArrayAssignedTextures()

DECLARATION

```
float MacroMicroVariationTexture(
    Texture2D tex,

    out int AssignedAVTextures[3],
    out int AssignedNSOTextures[3],
    out int AssignedEMTextures[3],
    out int AssignedBMTextures
)
```

PARAMETERS

Parameter	Description
tex	The texture to read

RETURNS

Output	Description
AssignedAVTextures	The assigned AVTextures, each value in the array is 32 bools
AssignedNSOTextures	The assigned NSOTextures, each value in the array is 32 bools
AssignedEMTextures	The assigned EMTextures, each value in the array is 32 bools
AssignedBMTextures	The assigned BMTextures (0-31)

DESCRIPTION

Loads the array assigned textures from a texture

14.4 Utilities

14.4.1 Utilities/BooleanCompression

Description

Used to compress and extract an array of boolean values in a single integer

AddCompressedValue()

DECLARATION

```
int AddCompressedValue(int CompressedValues, bool Value, int Index)
```

PARAMETERS

Parameter	Description
CompressedValues	Compressed int of bool values
Value	Value inserted into the compressedValue
Index	Index that the input value will overwrite

RETURNS

New compressed integer with the added value

DESCRIPTION

Overwrites a value at a given index into the input compressedValues

GetCompressedValue()

DECLARATION

```
bool GetCompressedValue(int CompressedValues, int Index)
```

PARAMETERS

Parameter	Description
CompressedValues	Compressed int of bool values
Index	Index of which bool value will be returned

RETURNS

Value at the given index

DESCRIPTION

Gets a compressed value at a given index from the input compressedValues

GetCompressedValue()

DECLARATION

```
bool GetCompressedValue(int CompressedValues[BOOLEAN_COMPRESSION_MAX_CHUNKS], int Index)
```

PARAMETERS

Parameter	Description
CompressedValues	Compressed int of bool values, each value in the array is 32 bools
Index	Index of which bool value will be returned

RETURNS

Value at the given index

DESCRIPTION

Gets a compressed value at a given index from the input compressedValues

---

Combine16BitInts()

DECLARATION

```
bool Combine16BitInts(int firstHalf, int secondHalf)
```

PARAMETERS

Parameter	Description
firstHalf	First half of the 32 bits
secondHalf	Second half of the 32 bits

RETURNS

The combined 32 bit integer

DESCRIPTION

Combines two halfs of a 32 bit integer

AddCompressedValue_float()

DECLARATION

```
void AddCompressedValue_float(int CompressedValues, bool Value, int Index, out int Out)
```

PARAMETERS

Parameter	Description
CompressedValues	Compressed int of bool values
Value	Value inserted into the compressedValue
Index	Index that the input value will overwrite

RETURNS

Output	Description
Out	New compressed integer with the added value

DESCRIPTION

Overwrites a value at a given index into the input compressedValues

Used in Add Compressed Value sub graph

GetCompressedValue_float()

DECLARATION

```
void GetCompressedValue_float(int CompressedValues, int Index, out bool Out)
```

PARAMETERS

Parameter	Description
CompressedValues	Compressed int of bool values
Index	Index of which bool value will be returned

RETURNS

Output	Description
Out	Value at the given index

DESCRIPTION

Gets a compressed value at a given index from the input compressedValues

Used in Get Compressed Value sub graph

14.4.2 Utilities/TextureUtilities

Description

Helpers for textures

Remap()

DECLARATION

```
float Remap(float In, float2 InMinMax, float2 OutMinMax)
```

PARAMETERS

Parameter	Description
In	The input value to remap
InMinMax	The current min max range the input value is inside
OutMinMax	The new min max range to remap the input value

RETURNS

Remapped value from InMinMax to OutMinMax

DESCRIPTION

Remaps the in value from InMinMax to OutMinMax

RotateUVDegrees()

DECLARATION

```
float2 RotateUVDegrees(float2 UV, float2 Center, float Rotation)
```

PARAMETERS

Parameter	Description
UV	The UVs to rotate
Center	The center point to rotate around
Rotation	The rotation angle in degrees

RETURNS

Rotated UVs

DESCRIPTION

Rotates the inputted UVs around the given Center by the given Rotation in degrees

---

UnpackNormalMap()

DECLARATION

```
float3 UnpackNormalMap(float4 PackedNormal, float Strength = 1.0)
```

PARAMETERS

Parameter	Description
PackedNormal	The packed normal map to unpack
Strength	Strength to apply to the normal map

RETURNS

The unpacked normal

DESCRIPTION

Unpacks the inputted normal and applies strength

14.5 Structs

14.5.1 Structs/RepetitionlessMaterialData

Description

Contains the data for a repetitionless material

Variables

Variable	Description
Settings1	x: Compressed Bools - 0 > NoiseEnabled - 1 > RandomiseNoiseScaling - 2 > RandomiseNoiseRotation - 3 > SmoothnessEnabled - 4 > VariationEnabled - 5 > PackedTexture - 6 > EmissionEnabled y: Compressed Bools - 0 > AlbedoAssigned - 1 > MetallicAssigned - 2 > SmoothnessAssigned - 3 > NormalAssigned - 4 > OcclussionAssigned - 5 > EmissionAssigned - 6 > VarationAssigned - 7 > PackedTextureAssigned z: Metallic w: Smoothness/Roughness
Settings2	x: NormalScale y: OcclussionStrength z: AlphaClipping w: NoiseAngleOffset
Settings3	x: NoiseScale y: VariationMode z: VariationOpacity w: VariationBrightness
Settings4	x: VariationSmallScale y: VariationMediumScale z: VariationLargeScale w: VariationNoiseStrength
Settings5	xy: NoiseScalingMinMax zw: NoiseRandomiseRotationMinMax
AlbedoTint	The albedo tint
EmissionColour	The emission colour
TilingOffset	The tiling offset: xy: Tiling, zw: Offset
VariationTO	**IF USING NOISE:** x: VariationNoiseScale zw: VariationNoiseOffset **IF USING TEXTURE:** xyzw: VariationTextureTO

14.6 TextureArrayEssentials

14.6.1 TextureArrayEssentials/TextureArrayUtilities

Description

Helpers for sampling texture arrays

GetIndexInArray()

DECLARATION

```
int GetIndexInArray(int TexturesAssignedCompressed[BOOLEAN_COMPRESSION_MAX_CHUNKS], int Index)
```

PARAMETERS

Parameter	Description
TexturesAssignedCompressed	The compressed assigned textures that contains which textures are assigned in the array, each value in the array is 32 bools
Index	The constant index of the texture in the array *(Doesnt care about how many textures are in the array, think of it as a constant within the max texture count)*

RETURNS

The actual index in the array from the constant index

DESCRIPTION

Gets the actual index in a texture array based off the assigned textures and constant index

GetIndexInArray_float()

DECLARATION

```
void GetIndexInArray_float(int TexturesAssignedCompressed, int Index, out int Out)
```

PARAMETERS

Parameter	Description
TexturesAssignedCompressed	The compressed assigned textures that contains which textures are assigned in the array
Index	The constant index of the texture in the array *(Doesnt care about how many textures are in the array, think of it as a constant within the max texture count)*

RETURNS

Output	Description
Out	The actual index in the array from the constant index

DESCRIPTION

Gets the actual index in a texture array based off the assigned textures and constant index

SampleArrayAtConstantIndex()

DECLARATION

```
float4 SampleArrayAtConstantIndex(
    Texture2DArray TextureArray,
    int TexturesAssignedCompressed[BOOLEAN_COMPRESSION_MAX_CHUNKS],
    int Index,
    float2 UV,
    float4 UnassignedColor,
    SamplerState SS
)
```

PARAMETERS

Parameter	Description
TextureArray	The texture array to sample
TexturesAssignedCompressed	The compressed assigned textures that contains which textures are assigned in the array, each value in the array is 32 bools
Index	The constant index of the texture in the array *(Doesnt care about how many textures are in the array, think of it as a constant within the max texture count)*
UV	UV used for sampling the texture
UnassignedColor	The output colour if no texture is found at the inputted Index
SS	Sampler state used for sampling the texture

RETURNS

Output	Description
Out	The output colour

DESCRIPTION

Samples a texture array based on a constant index

SampleArrayAtConstantIndex_float()

DECLARATION

```
void SampleArrayAtConstantIndex_float(
    Texture2DArray TextureArray,
    int TexturesAssignedCompressed,
    int Index,
    float2 UV,
    float4 UnassignedColor,
    SamplerState SS,
    out float4 Out
)
```

PARAMETERS

Parameter	Description
TextureArray	The texture array to sample
TexturesAssignedCompressed	The compressed assigned textures that contains which textures are assigned in the array
Index	The constant index of the texture in the array *(Doesnt care about how many textures are in the array, think of it as a constant within the max texture count)*
UV	UV used for sampling the texture
UnassignedColor	The output colour if no texture is found at the inputted Index
SS	Sampler state used for sampling the texture

RETURNS

Output	Description
Out	The output colour

DESCRIPTION

Samples a texture array based on a constant index

15. Scripting API

15.1 Runtime

15.1.1 Runtime.RepetitionlessTerrain

Description

Handles Repetitionless materials interfacing with a terrain, automatically updating terrain textures and syncing the terrain layers to the material

Variables

Variable	Description
MainMaterial	The main material set in the inspector
MaterialInstance	The instance of the material the terrain is using
AutoSaveTextures	If modifying the terrain layers will automatically update the material
Terrain	The terrain being used
ParentTerrain	The parent terrain that is handling this terrains material
OnTerrainLayersChanged	Callback when the terrain layers are changed
OnMaterialAssigned	Callback when the material is changed
OnMaterialTexturesUpdated	Callback when the material textures are updated

---

UpdateTerrainMaterial(Material, bool)

DECLARATION

```
public void UpdateTerrainMaterial(Material material, bool assignMaterial = true)
```

PARAMETERS

Parameter	Description
material	The material that will be instanced
assignMaterial	If the material instance should be assigned to the terrain

DESCRIPTION

Creates a new material instance and updates the terrain

AssignMaterialInstance()

DECLARATION

```
public void AssignMaterialInstance()
```

DESCRIPTION

Assigns the currently used material instance to the terrain

UpdateMaterialTerrainTextures()

DECLARATION

```
public void UpdateMaterialTerrainTextures()
```

DESCRIPTION

Updates the terrain textures on the material instance

UpdateLayersCount()

DECLARATION

```
public void UpdateLayersCount()
```

DESCRIPTION

Updates the layer cound on the material instance

UpdateControlTextures()

DECLARATION

```
public void UpdateControlTextures()
```

DESCRIPTION

Updates the control textures on the material instance

UpdateParentCallback(RepetitionlessTerrain)

DECLARATION

```
public void UpdateParentCallback(RepetitionlessTerrain newParent)
```

PARAMETERS

Parameter	Description
newParent	The new parent being subscribed to

DESCRIPTION

Updates the parent callbacks

Assumes this is called from the editor before the current parent is updated

---

ParentMaterialChanged(Material)

DECLARATION

```
public void ParentMaterialChanged(Material material)
```

PARAMETERS

Parameter	Description
material	The new material to use

DESCRIPTION

Updates the terrain material and its textures

Called by the parent when its material has changed

ParentMaterialTexturesUpdated()

DECLARATION

```
public void ParentMaterialTexturesUpdated()
```

DESCRIPTION

Updates the material terrain textures

Called by the parent when its material has changed

SetupNewTerrainNeighbour(Terrain)

DECLARATION

```
public void SetupNewTerrainNeighbour(Terrain newNeighbour)
```

PARAMETERS

Parameter	Description
newNeighbour	The new terrain to setup

DESCRIPTION

Sets up a terrain neighbour with a RepetitionlessTerrain component with this terrain as its parent

Called via the editor when a terrain is created beside this one

15.1.2 Compression

Runtime.Compression.BooleanCompression

DESCRIPTION

Used to compress and extract an array of boolean values in a single integer

VARIABLES

Variable	Description
MAX_VALUES	The max values that can be stored in one compressed integer (32-bit integer)

COMPRESSVALUES(PARAMS BOOL[])

Declaration

```
public static int CompressValues(params bool[] values)
```

Parameters

Parameter	Description
values	Bools that will be compressed

Returns

Compressed int of bools

Description

Compresses the input array of bools into an int

GETVALUES(INT, INT)

Declaration

```
public static bool[] GetValues(int compressedValues, int valueCount)
```

Parameters

Parameter	Description
compressedValues	Compressed int of bool values
valueCount	The total amount of values stored in the compressedValues

Returns

Array of all the values stored in the input compressedValues

Description

Retrieves all the boolean values stored in a compressed int

ADDVALUE(INT, INT, BOOL)

Declaration

```
public static int AddValue(int compressedValues, int index, bool value)
```

Parameters

Parameter	Description
compressedValues	Compressed int of bool values
index	Index that the input value will overwrite
value	Value inserted into the compressedValue

Returns

New compressed integer with the added value

Description

Overwrites a value at a given index into the input compressedValues

---

GETVALUE(INT, INT)

Declaration

```
public static bool GetValue(int compressedValues, int index)
```

Parameters

Parameter	Description
compressedValues	Compressed int of bool values
index	Index of which bool value will be returned

Returns

Value at the given index

Description

Gets a compressed value at a given index from the input compressedValues

---

SPLIT32BITINT(INT)

Declaration

```
public static (ushort, ushort) Split32BitInt(int value)
```

Parameters

Parameter	Description
value	The integer to split

Returns

Item1: First half
Item2: Second half

Description

Splits a 32 bit integer into two 16 bit integers

COMBINE16BITINTS(USHORT, USHORT)

Declaration

```
public static int Combine16BitInts(ushort firstHalf, ushort secondHalf)
```

Parameters

Parameter	Description
firstHalf	The first integer
secondHalf	The second integer

Returns

The combined 32 bit integer

Description

Combines two 16 bit integers into a 32 bit integer

15.1.3 Variables

Runtime.Variables.ESurfaceType

DESCRIPTION

Available surface types for Repetitionless Materials

MEMBERS

Member	Description
Opaque	Opaque surface with no transparency
Cutout	TransparentCutout surface (Opaque with alpha clipping)
Transparent	Transparent surface

Runtime.Variables.EUVSpace

DESCRIPTION

Available UV spaces

MEMBERS

Member	Description
Local	Local uvs from the object
World	World uvs from the world position

Runtime.Variables.ENoiseQuality

DESCRIPTION

Available noise qualities

MEMBERS

Member	Description
High	Dynamically creates the noise on the fly Required to adjust the Noise Angle Offset
Medium	Uses the 4k pre-rendered noise texture
Low	Uses the 1k pre-rendered noise texture

Runtime.Variables.EVariationType

DESCRIPTION

Available variation texture types for Repetitionless Materials

MEMBERS

Member	Description
PerlinNoise	Procedurally generated with perlin noise
SimplexNoise	Procedurally generated with simplex noise
CustomTexture	Uses a custom input texture

Runtime.Variables.EMaskType

DESCRIPTION

Available mask texture types for Repetitionless Materials

MEMBERS

Member	Description
PerlinNoise	Procedurally generated with perlin noise
SimplexNoise	Procedurally generated with simplex noise
CustomTexture	Uses a custom input texture
VertexColour	Uses a defined colour from the vertex data

Runtime.Variables.EDistanceBlendMode

DESCRIPTION

Available distance blend modes for Repetitionless Materials

MEMBERS

Member	Description
TilingOffset	Blends into the same material with a different tiling and offset
Material	Blends into a different material

Runtime.Variables.EVertexColourBlendMode

DESCRIPTION

Available blend modes for the vertex colour

MEMBERS

Member	Description
False	Doesnt use the vertex colour
Multiply	Multiplies the vertex colour with the output
Additive	Adds the vertex colour to the output
Subtractive	Subtracts the vertex colour to the output
Overwrite	Overwrites the output colour

Runtime.Variables.ELayerMode

DESCRIPTION

Available options for how layers are used

MEMBERS

Member	Description
TerrainLayers	Uses automatically synced terrain textures and its terrain layers to assign textures and settings to each layer
ControlTextures	Uses manually set textures to specify where each layer is

Runtime.Variables.ERenderPipeline

DESCRIPTION

Available render pipelines

MEMBERS

Member	Description
Builtin	Default Built-In Pipeline
URP	Universal Render Pipeline
HDRP	High Definition Render Pipeline
Unknown	Unknown Pipeline

Runtime.Variables.RepetitionlessMaterialData

DESCRIPTION

The material data for a repetitionless material

VARIABLES

Variable	Description
NoiseEnabled	If noise is enabled
RandomiseNoiseScaling	If each cell will randomly scale
RandomiseNoiseRotation	If each cell will randomly rotate
SmoothnessEnabled	If smoothness or roughness is used
VariationEnabled	If texture variation is enabled
PackedTexture	If a packed texture is used
EmissionEnabled	If emission is enabled
AlbedoAssigned	If the albedo texture is assigned
MetallicAssigned	If the metallic texture is assigned
SmoothnessAssigned	If the smoothness texture is assigned
NormalAssigned	If the normal texture is assigned
OcclussionAssigned	If the occlussion texture is assigned
EmissionAssigned	If the emission texture is assigned
VariationAssigned	If the variation texture is assigned
PackedTextureAssigned	If the packed texture is assigned
AlbedoTint	The albedo tint
EmissionColour	The emission colour
TilingOffset	The tiling offset: xy: Tiling, zw: Offset
Metallic	The metallic value
SmoothnessRoughness	The smoothness/roughness value
NormalScale	The normal scale
OcclussionStrength	The occlussion strength
AlphaClipping	The alpha clipping value
NoiseAngleOffset	The angle offset for the noise
NoiseScale	The scale of the noise
NoiseScalingMinMax	The random noise scaling min max
NoiseRandomiseRotationMinMax	The random noise rotation min max
VariationMode	The variation mode used
VariationOpacity	The variation opacity
VariationBrightness	The variation brightness
VariationSmallScale	The variation small sample scale
VariationMediumScale	The variation medium sample scale
VariationLargeScale	The variation large sample scale
VariationNoiseStrength	The variation noise strength

Variable	Description
VariationNoiseScale	The variation noise scale
VariationNoiseOffset	The variation noise offset
VariationTextureTO	The variation texture tiling offset: xy: Tiling, zw: Offset

Runtime.Variables.RepetitionlessMaterialDataCompressed

DESCRIPTION

The compressed material data that will be passed to the shader for a repetitionless material

MEMBERS

Member	Description
Settings1	x: Compressed Bools 0 > NoiseEnabled 1 > RandomiseNoiseScaling 2 > RandomiseNoiseRotation 3 > SmoothnessEnabled 4 > VariationEnabled 5 > PackedTexture 6 > EmissionEnabled y: Compressed Bools 0 > AlbedoAssigned 1 > MetallicAssigned 2 > SmoothnessAssigned 3 > NormalAssigned 4 > OcclussionAssigned 5 > EmissionAssigned 6 > VarationAssigned 7 > PackedTextureAssigned z: Metallic w: Smoothness/Roughness
Settings2	x: NormalScale y: OcclussionStrength z: AlphaClipping w: NoiseAngleOffset
Settings3	x: NoiseScale y: VariationMode z: VariationOpacity w: VariationBrightness
Settings4	x: VariationSmallScale y: VariationMediumScale z: VariationLargeScale w: VariationNoiseStrength
Settings5	xy: NoiseScalingMinMax zw: NoiseRandomiseRotationMinMax
AlbedoTint	The albedo tint
EmissionColour	The emission colour
TilingOffset	The tiling offset: xy: Tiling, zw: Offset
VariationTO	**IF USING NOISE:** x: VariationNoiseScale zw: VariationNoiseOffset **IF USING TEXTURE:** xyzw: VariationTextureTO

Runtime.Variables.RepetitionlessLayerData

DESCRIPTION

The layer data for a repetitionless material

VARIABLES

Variable	Description
BaseMaterialData	The base material data
FarMaterialData	The far material data
BlendMaterialData	The blend material data
GlobalTilingOffset	When packing this will update all the materials tiling offset
DistanceBlendEnabled	If distance blend is enabled
DistanceBlendMode	The distance blend mode used
DistanceBlendMinMax	The distance blend min max
MaterialBlendEnabled	If material blend is enabled
BlendMaskAssigned	If the blend mask texture is assigned
OverrideDistanceBlend	If the blend material will be drawn at a distance instead of the distance material
OverrideDistanceBlendTO	If the blend material at a distance will have a different tiling offset
BlendMaskType	The blend mask type used
BlendMaskDistanceTO	The blend mask distance tiling offset: xy: Tiling, zw: Offset
BlendMaskOpacity	The blend mask opacity
BlendMaskStrength	The blend mask strength
BlendMaskNoiseScale	The blend mask noise scale
BlendMaskNoiseOffset	The blend mask noise offset
BlendMaskTextureTO	The blend mask texture tiling offset: xy: Tiling, zw: Offset
BlendMaskVertexColour	The colour used as a mask in the vertex colour
BlendMaskVertexColourThreshold	The smoothstep min max for the colour

Runtime.Variables.RepetitionlessLayerDataCompressed

DESCRIPTION

The compressed layer data that will be passed to the shader fro a repetitionless material

MEMBERS

Member	Description
BaseMaterialData	The compressed base material data
FarMaterialData	The compressed far material data
BlendMaterialData	The compressed blend material data
DistanceBlendSettings	x: DistanceBlendEnabled y: DistanceBlendMode zw: DistanceBlendMinMax
BlendMaskDistanceTO	The blend mask distance tiling offset: xy: Tiling, zw: Offset
MaterialBlendSettings	x: Compressed Bools 0 > MaterialBlendEnabled 1 > BlendMaskAssigned 2 > OverrideDistanceBlend 3 > OverrideDistanceBlendTO y: BlendMaskType z: BlendMaskOpacity w: BlendMaskStrength
MaterialBlendMaskTO	**IF USING NOISE MASK:** x: BlendMaskNoiseScale zw: BlendMaskNoiseOffset **IF USING TEXTURE MASK:** xyzw: BlendMaskTextureTO **IF USING VERTEX COLOUR MASK:** xyzw: BlendMaskVertexColour
MaterialBlendMaskExtraSettings	**IF USING VERTEX COLOUR:** xy: BlendMaskVertexColourThreshold

15.2 Editor

15.2.1 Data

Editor.Data.MaterialDataManager

DESCRIPTION

`Unity Editor Only`

Manages the data folder for a repetitionless material

Stores the data in a folder accompanying the material

VARIABLES

Variable	Description
Material	The material this is handling data for
MaterialDataManager(Material)	MaterialDataManager Constructor
MaterialDataManager(Object)	Gets a data manager from the path of an asset inside the data folder If the main material cannot be found, Material will be null

GENERATEFOLDERNAME(STRING)

Declaration

```
public static string GenerateFolderName(string prefix)
```

Parameters

Parameter	Description
prefix	The input folder name

Returns

The new folder name

Description

Adds the folder suffix to the input name

DATAFOLDERPARENTPATH()

Declaration

```
public string DataFolderParentPath()
```

Returns

The parent folder path

Description

Gets the path of the data folder parent

DATAFOLDERNAME()

Declaration

```
public string DataFolderName()
```

Returns

The data folder name

Description

Gets the data folder name

DATAFOLDERPATH()

Declaration

```
public string DataFolderPath()
```

Returns

The data folder path

Description

Gets the data folder path

CREATEASSET(OBJECT, STRING, BOOL)

Declaration

```
public void CreateAsset(Object asset, string fileName, bool overwrite = false)
```

Parameters

Parameter	Description
asset	The asset to create
fileName	The file name of the asset
overwrite	If it will overwrite an asset with the same name if it exists

Description

Creates an asset in the data folder

LOADASSET(STRING)

Declaration

```
public T LoadAsset<T>(string fileName) where T : Object
```

Parameters

Parameter	Description
fileName	The filename of the asset

Returns

The asset

Description

Loads an asset from the data folder

ASSETEXISTS(STRING)

Declaration

```
public bool AssetExists(string fileName)
```

Parameters

Parameter	Description
fileName	The file name to check

Returns

If the asset exists

Description

Checks if an asset exists in the data folder with the given name

---

DELETEASSET(STRING)

Declaration

```
public void DeleteAsset(string fileName)
```

Parameters

Parameter	Description
fileName	The asset file name

Description

Deletes an asset in the data folder with the given name if it exists

---

Editor.Data.RepetitionlessMaterialDataSO

DESCRIPTION

`Unity Editor Only`

Stores the material properties for a Repetitionless material

Creates and manages a texture storing these properties that will be passed to the shader

VARIABLES

Variable	Description
PROPERTIES_TEXTURE_PROP_NAME	The properties texture material property
Data	The data for the material Do not update this in the scriptable object, do it in the material inspector
GlobalTilingOffset	The global tiling offset that is being used for all the data
OnExternalDataChanged	This is called by other classes when they have changed properties and manually called this It is not called when regular properties are changed by this class

---

INIT(INT)

Declaration

```
public void Init(int layerCount)
```

Parameters

Parameter	Description
layerCount	The max amount of terrain layers that will be used

Description

Initializes this with a new set of data

SAVE()

Declaration

```
public void Save()
```

Description

Saves this object

SETNOISEQUALITY(ENOISEQUALITY)

Declaration

```
public void SetNoiseQuality(ENoiseQuality noiseQuality)
```

Parameters

Parameter	Description
noiseQuality	The new quality to use

Description

Sets the noise quality

SETGLOBALTILINGOFFSET(VECTOR4)

Declaration

```
public void SetGlobalTilingOffset(Vector4 tilingOffset)
```

Parameters

Parameter	Description
tilingOffset	The new tiling offset to use

Description

Sets the global tiling and offset for every layer

CALLONEXTERNALDATACHANGED()

Declaration

```
public void CallOnExternalDataChanged()
```

Description

Invokes the OnForcedDataChanged callback

UPDATEMATERIALTEXTURE(MATERIAL, INT)

Declaration

```
public void UpdateMaterialTexture(Material material, int layerIndex)
```

Parameters

Parameter	Description
material	The material that will have its texture property updated
layerIndex	The layer that will be updated

Description

Updates the properties texture with the data saved in this object

---

UPDATEMATERIALTEXTURE(MATERIAL, STRING, INT)

Declaration

```
public void UpdateMaterialTexture(Material material, string texturePropertyName, int layerIndex)
```

Parameters

Parameter	Description
material	The material that will have its texture property updated
texturePropertyName	The name of the properties texture property
layerIndex	The layer that will be updated

Description

Updates the properties texture with the data saved in this object

---

UPDATEMATERIALTEXTURE(MATERIALPROPERTY, INT)

Declaration

```
public void UpdateMaterialTexture(MaterialProperty property, int layerIndex)
```

Parameters

Parameter	Description
property	The properties texture property that will be updated
layerIndex	The layer that will be updated

Description

Updates the properties texture with the data saved in this object

---

GETMATERIALDATA(INT, INT)

Declaration

```
public RepetitionlessMaterialData GetMaterialData(int layerIndex, int materialIndex)
```

Parameters

Parameter	Description
layerIndex	The layer that the material is in
materialIndex	The index of the material: 0: Base, 1: Far, 2: Blend

Returns

The properties for a specific material

Description

Gets the properties for a specific material

UPDATEASSIGNEDTEXTURES(MATERIAL, REPETITIONLESSTEXTUREDATASO, INT, INT)

Declaration

```
public void UpdateAssignedTextures(Material material, RepetitionlessTextureDataSO textureData, int layerIndex, int materialIndex)
```

Parameters

Parameter	Description
material	The material that will have its texture property updated
textureData	The texture data that assigned textures will be read from
layerIndex	The layer that the material is in
materialIndex	The index of the material: 0: Base, 1: Far, 2: Blend

Description

Updates the assigned textures and the property texture based on a texture data

Editor.Data.RepetitionlessTextureDataSO

DESCRIPTION

`Unity Editor Only`

Stores the textures for a RepetitionlessMaterial

Handles drawing and updating the texture arrays

VARIABLES

Variable	Description
TEXTURE_PROP_NAME	The material property name of the array assigned textures texture
LayersTextureData	The data for the textures
PackedColourSpace	Which colour space the textures were previously packed in Used when switching colour spaces to determine which textures need to be repacked
AVTexturesDrawer	The AVTextures drawer Not serialized, needs to be setup with SetupTextureDrawers for each session using this
NSOTexturesDrawer	The NSOTextures drawer Not serialized, needs to be setup with SetupTextureDrawers for each session using this
EMTexturesDrawer	The EMTextures drawer Not serialized, needs to be setup with SetupTextureDrawers for each session using this
BMTexturesDrawer	The BMTextures drawer Not serialized, needs to be setup with SetupTextureDrawers for each session using this
OnDataChanged	Callback for when any of the textures are changed

---

INIT(INT)

Declaration

```
public void Init(int layersCount)
```

Parameters

Parameter	Description
layersCount	The max amount of terrain layers that will be used

Description

Initializes this with a new set of texture data

SETUPLAYER(INT)

Declaration

```
public void SetupLayer(int layerIndex)
```

Parameters

Parameter	Description
layerIndex	The layer index to setup

Description

Initializes a repetitionless layer with new texture data

SETUPMATERIAL(REF MATERIALTEXTUREDATA)

Declaration

```
public void SetupMaterial(ref RepetitionlessTextureDataSO.MaterialTextureData data)
```

Parameters

Parameter	Description
data	A reference to the material texture data being initialized

Description

Initializes a repetitionless material with new texture data

SAVE()

Declaration

```
public void Save()
```

Description

Saves this object

UPDATEASSIGNEDTEXTURESTEXTURE()

Declaration

```
public void UpdateAssignedTexturesTexture()
```

Description

Updates the array assigned textures texture

UPDATEASSIGNEDTEXTURESTEXTURE(MATERIALPROPERTY)

Declaration

```
public void UpdateAssignedTexturesTexture(MaterialProperty property)
```

Parameters

Parameter	Description
property	The assigned textures property that will be updated

Description

Updates the array assigned textures texture

SETUPTEXTUREDRAWERS()

Declaration

```
public void SetupTextureDrawers()
```

Description

Initializes the texture drawers

These are not serialized and this must be called for each session using them

---

GETMATERIALTEXTUREDATA(INT, INT)

Declaration

```
public ref RepetitionlessTextureDataSO.MaterialTextureData GetMaterialTextureData(int layerIndex, int materialIndex)
```

Parameters

Parameter	Description
layerIndex	The layer to retrieve data from
materialIndex	The material to retrieve data from: 0: Base material, 1: Far material, 2: Blend material

Returns

Description

Gets a reference to the material texture data saved in this object

---

GETTEXTUREDATA(REF MATERIALTEXTUREDATA, INT)

Declaration

```
public ref TexturePacker.TextureData[] GetTextureData(ref RepetitionlessTextureDataSO.MaterialTextureData materialData, int texturesIndex)
```

Parameters

Parameter	Description
materialData	The material data to get the texture data from
texturesIndex	The index of the textures that will be used: 0: AV, 1: NSO, 2: EM, 3: BM

Returns

A reference to the texture data

Description

Gets a reference to the texture data saved in a material data

---

GETTEXTUREDATA(INT, INT, INT)

Declaration

```
public ref TexturePacker.TextureData[] GetTextureData(int layerIndex, int materialIndex, int texturesIndex)
```

Parameters

Parameter	Description
layerIndex	The layer to retrieve data from
materialIndex	The index of the material: 0: Base, 1: Far, 2: Blend
texturesIndex	The index of the textures that will be used: 0: AV, 1: NSO, 2: EM, 3: BM

Returns

A reference to the texture data

Description

Gets a reference to the texture data saved in a material data

---

UPDATEPACKEDTEXTURE(INT, INT, BOOL)

Declaration

```
public void UpdatePackedTexture(int layerIndex, int materialIndex, bool enabled)
```

Parameters

Parameter	Description
layerIndex	The layer to update
materialIndex	The index of the material: 0: Base, 1: Far, 2: Blend
enabled	The new state of the packed texture

Description

Updates the packed texture data and the associated arrays

RepetitionlessTextureDataSO Variables

EDITOR.DATA.REPETITIONLESSTEXTUREDATASO.MATERIALTEXTUREDATA

Description

`Unity Editor Only`

Contains the texture data for a repetitionless material

Members

Member	Description
AVTextures	The AVTextures: Albedo (rgb), Variation (a)
NSOTextures	The NSOTextures: Normal (rg), Smoothness (b), Occlussion (a)
EMTextures	The EMTextures: Emission (rgb), Metallic (a)

EDITOR.DATA.REPETITIONLESSTEXTUREDATASO.LAYERTEXTUREDATA

Description

`Unity Editor Only`

Contains the texture data for a repetitionless layer

Variables

Variable	Description
BaseMaterialTextures	The base material textures
FarMaterialTextures	The far material textures
BlendMaterialTextures	The blend material textures
BlendMaskTexture	The blend mask texture for the layer Stored in an array to make it easier to pass into the texture drawer

Editor.Data.RepetitionlessTerrainDataSO

DESCRIPTION

`Unity Editor Only`

Stores the terrain data for a Repetitionless material

VARIABLES

Variable	Description
TerrainLayers	The terrain layers linked to this material
AutoUpdateMaxLayers	If the max layers is updated automatically with terrain layers
AutoSyncLayers	Toggles if textures and settings are automatically saved and loaded to and from the terrain layers

---

SAVE()

Declaration

```
public void Save()
```

Description

Saves this object

UPDATETERRAINLAYERS(TERRAINLAYER[])

Declaration

```
public void UpdateTerrainLayers(TerrainLayer[] layers)
```

Parameters

Parameter	Description
layers	The new terrain layers to use

Description

Updates the terrain layers, overwrites the currently set with the inputted layers

---

CLEARTERRAINLAYERS()

Declaration

```
public void ClearTerrainLayers()
```

Description

Clears the terrain layers

UPDATELAYERMATERIALDATA(TERRAINLAYER, BOOL)

Declaration

```
public void UpdateLayerMaterialData(TerrainLayer terrainLayer, bool forceUpdate = false)
```

Parameters

Parameter	Description
terrainLayer	The terrain layer to update
forceUpdate	Ignores the AutoSyncLayers setting and updates the data anyway

Description

Updates the layer linked to the inputted terrain layer on the material

---

REMOVEUNUSEDLAYERTEXTURES()

Declaration

```
public void RemoveUnusedLayerTextures()
```

Description

Removes any unused terrain layers for a material

Editor.Data.RepetitionlessDataPacker

DESCRIPTION

`Unity Editor Only`

Compresses repetitionless material data

VECTOR4TOCOLOUR(VECTOR4)

Declaration

```
public static Color Vector4ToColour(Vector4 vector)
```

Parameters

Parameter	Description
vector	The vector to convert

Returns

The colour

Description

Converts a vector4 to a colour

VECTOR3TOCOLOUR(VECTOR3)

Declaration

```
public static Color Vector3ToColour(Vector3 vector)
```

Parameters

Parameter	Description
vector	The vector to convert

Returns

The colour

Description

Converts a vector3 to a colour

UPDATECOMPRESSEDMATERIALDATA(REF REPETITIONLESSMATERIALDATACOMPRESSED, REPETITIONLESSMATERIALDATA, VECTOR4, BOOL)

Declaration

```
public static int UpdateCompressedMaterialData(ref RepetitionlessMaterialDataCompressed compressedData, RepetitionlessMaterialData newData, Vector4
globalTilingOffset, bool updateAll = false)
```

Parameters

Parameter	Description
compressedData	Reference to the compressed data that will be updated
newData	The new data to compress
globalTilingOffset	The global tiling offset for the material
updateAll	If all the values will be updated if changed If disabled only the first changed value will update

Returns

Returns the changed variable index in the struct
If updateAll is enabled it will return the default -1

Description

Updates changed values in the compressed material data based on the given material data

---

UPDATECOMPRESSEDLAYERDATA(REF REPETITIONLESSLAYERDATACOMPRESSED, REPETITIONLESSLAYERDATA, BOOL)

Declaration

```
public static int UpdateCompressedLayerData(ref RepetitionlessLayerDataCompressed compressedData, RepetitionlessLayerData newData, bool updateAll = false)
```

Parameters

Parameter	Description
compressedData	Reference to the compressed data that will be updated
newData	The new data to compress
updateAll	If all the values will be updated if changed If disabled only the first changed value will update

Returns

Returns the changed variable index in the struct, incrementing for each material
If updateAll is enabled it will return the default -1

Description

Updates changed values in the compressed layer data based on the given layer data

---

GETMATERIALFIELDCOLOUR(REPETITIONLESSMATERIALDATACOMPRESSED, INT)

Declaration

```
public static Color? GetMaterialFieldColour(RepetitionlessMaterialDataCompressed compressedData, int compressedFieldIndex)
```

Parameters

Parameter	Description
compressedData	The compressed data to get the value from
compressedFieldIndex	The index of the compressed field to get: 0: Settings1 1: Settings2 2: Settings3 3: Settings4 4: Settings5 5: AlbedoTint 6: EmissionColour 7: TilingOffset 8: VariationTO

Returns

The nullable field colour, will be null if compressedFieldIndex was outside the range of values

Description

Gets a material field colour based on the given index

GETLAYERFIELDCOLOUR(REPETITIONLESSLAYERDATACOMPRESSED, INT)

Declaration

```
public static Color? GetLayerFieldColour(RepetitionlessLayerDataCompressed compressedData, int compressedFieldIndex)
```

Parameters

Parameter	Description
compressedData	The compressed data to get the value from
compressedFieldIndex	The index of the compressed field to get: 0-8: Base Material 9-17: Far Material 18-26: Blend Material 27: DistanceBlendSettings 28: BlendMaskDistanceTO 29: MaterialBlendSettings 30: MaterialBlendMaskTO

Returns

The nullable field colour, will be null if compressedFieldIndex was outside the range of values

Description

Gets a layer field colour based on the given index

15.2.2 Materials

Editor.Materials.RepetitionlessMaterialCreator

DESCRIPTION

`Unity Editor Only`

Used to create new repetitionless materials

CREATEMATERIAL(ERENDERPIPELINE, STRING, STRING, BOOL)

Declaration

```
public static MaterialDataObjects CreateMaterial(ERenderPipeline pipeline, string folderPath, string fileName = "RepetitionlessMaterial.mat", bool ping =
true)
```

Parameters

Parameter	Description
pipeline	The materials render pipeline
folderPath	The folder to save the material to
fileName	The file name for the material Must include the extension .mat
ping	If the material will be selected and pinged in the project window after creation

Returns

The data objects for the created material

Description

Creates a repetitionless material

CREATEMATERIAL(STRING, STRING, BOOL)

Declaration

```
public static MaterialDataObjects CreateMaterial(string folderPath, string fileName = "RepetitionlessMaterial.mat", bool ping = true)
```

Parameters

Parameter	Description
folderPath	The folder to save the material to
fileName	The file name for the material Must include the extension .mat
ping	If the material will be selected and pinged in the project window after creation

Returns

The data objects for the created material

Description

Creates a repetitionless material

Uses the currently selected render pipeline

CREATEMATERIALATCURRENTFOLDER(BOOL)

Declaration

```
public static MaterialDataObjects CreateMaterialAtCurrentFolder(bool ping = true)
```

Parameters

Parameter	Description
ping	If the material will be selected and pinged in the project window after creation

Returns

The data objects for the created material

Description

Creates a repetitionless material

Uses the currently selected render pipeline

Saves to the currently opened folder in the project window

CREATETERRAINMATERIAL(ERENDERPIPELINE, STRING, STRING, BOOL)

Declaration

```
public static MaterialDataObjects CreateTerrainMaterial(ERenderPipeline pipeline, string folderPath, string fileName = "RepetitionlessLayeredMaterial.mat",
bool ping = true)
```

Parameters

Parameter	Description
pipeline	The materials render pipeline
folderPath	The folder to save the material to
fileName	The file name for the material Must include the extension .mat
ping	If the material will be selected and pinged in the project window after creation

Returns

The data objects for the created material

Description

Creates a repetitionless terrain material

CREATETERRAINMATERIAL(STRING, STRING, BOOL)

Declaration

```
public static MaterialDataObjects CreateTerrainMaterial(string folderPath, string fileName = "RepetitionlessLayeredMaterial.mat", bool ping = true)
```

Parameters

Parameter	Description
folderPath	The folder to save the material to
fileName	The file name for the material Must include the extension .mat
ping	If the material will be selected and pinged in the project window after creation

Returns

The data objects for the created material

Description

Creates a repetitionless terrain material

Uses the currently selected render pipeline

CREATETERRAINMATERIALATCURRENTFOLDER(BOOL)

Declaration

```
public static MaterialDataObjects CreateTerrainMaterialAtCurrentFolder(bool ping = true)
```

Parameters

Parameter	Description
ping	If the material will be selected and pinged in the project window after creation

Returns

The data objects for the created material

Description

Creates a repetitionless terrain material

Uses the currently selected render pipeline

Saves to the currently opened folder in the project window

SETUPMATERIAL(MATERIAL, INT, ACTION)

Declaration

```
public static MaterialDataObjects SetupMaterial(Material mat, int maxLayers, Action<RepetitionlessMaterialDataSO> onPropertiesCreatedCallback = null)
```

Parameters

Parameter	Description
mat	The material to setup
maxLayers	The max amount of layers that can be used
onPropertiesCreatedCallback	Callback for when the material properties are created

Returns

The data objects for the material

Description

Loads or Creates the data objects for a repetitionless material

Editor.Materials.RepetitionlessMaterialConverter

DESCRIPTION

`Unity Editor Only`

Used to convert lit materials to repetitionless materials

CONVERTMATERIAL(MATERIAL)

Declaration

```
public static Material ConvertMaterial(Material material)
```

Parameters

Parameter	Description
material	The material to convert

Returns

The created material

Description

Convert a material to a repetitionless material

Saves the material next to the original

CONVERTMATERIALS(MATERIAL[], BOOL)

Declaration

```
public static void ConvertMaterials(Material[] materials, bool selectConverted = true)
```

Parameters

Parameter	Description
materials	The materials to convert
selectConverted	If the materials will be selected in the project window after creation

Description

Convert multiple materials to repetitionless materials

Saves the materials next to the originals

Editor.Materials.MaterialDataObjects

DESCRIPTION

`Unity Editor Only`

Holds all the data related to a repetitionless material

MEMBERS

Member	Description
Material	The material
DataManager	The data manager
TextureDataSO	The texture data scriptable object
MaterialDataSO	The material data scriptable object

15.2.3 GUIUtilities

Editor.GUIUtilities.GUIUtilities

DESCRIPTION

`Unity Editor Only`

Many GUI functions to make inspector creation easier without MaterialProperties

VARIABLES

Variable	Description
LINE_HEIGHT	The height that one line takes in the inspector
LINE_SPACING	The height in-between two lines in the inspector
FOLDOUT_INDENT	The amount of left padding for foldouts
BoldHeaderLargeStyle	Style used for large headers
MajorToggleButtonStyle	Style for major toggle buttons
BoldFoldoutStyle	Style used for foldout headers

GETLINERECT(FLOAT)

Declaration

```
public static Rect GetLineRect(float heightOverride = 20)
```

Parameters

Parameter	Description
heightOverride	Default **20 (LINE_HEIGHT + LINE_SPACING)** Sets the height of the output rect

Returns

The rect of a single line

Description

Gets the rect of a single line, updating the GUILayout accordingly

DRAWHEADERLABELLARGE(STRING)

Declaration

```
public static void DrawHeaderLabelLarge(string text)
```

Parameters

Parameter	Description
text	The text to show

Description

Draws a Large Header Label

DRAWMAJORTOGGLEBUTTON(MATERIALPROPERTY, STRING)

Declaration

```
public static bool DrawMajorToggleButton(MaterialProperty property, string label)
```

Parameters

Parameter	Description
property	The material property to be used and modified for the toggle state
label	The text to show inside the button

Returns

The button toggled state

Description

Draws a large toggle button that uses and updates the inputted property

Displays red when disabled, green when enabled

DRAWMAJORTOGGLEBUTTON(BOOL, STRING)

Declaration

```
public static bool DrawMajorToggleButton(bool enabled, string label)
```

Parameters

Parameter	Description
enabled	Toggle state if the button is enabled
label	The text to show inside the button

Returns

The button toggled state

Description

Draws a large toggle button using an input toggle state value

Displays red when disabled, green when enabled

DRAWTEXTURE2DARRAY(TEXTURE2DARRAY, GUICONTENT)

Declaration

```
public static Texture2DArray DrawTexture2DArray(Texture2DArray array, GUIContent content)
```

Parameters

Parameter	Description
array	The input Texture2DArray
content	The GUIContent for the field

Returns

The Texture2DArray input into the inspector

Description

Draws a Texture2DArray field

DRAWTEXTURE2DARRAY(RECT, TEXTURE2DARRAY, GUICONTENT)

Declaration

```
public static Texture2DArray DrawTexture2DArray(Rect rect, Texture2DArray array, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
array	The input Texture2DArray
content	The GUIContent for the field

Returns

The Texture2DArray input into the inspector

Description

Draws a Texture2DArray field

DRAWTEXTURE(TEXTURE2D, GUICONTENT)

Declaration

```
public static Texture2D DrawTexture(Texture2D texture, GUIContent content)
```

Parameters

Parameter	Description
texture	The input texture
content	The GUIContent for the field

Returns

The texture input into the inspector

Description

Draws a single texture field

DRAWTEXTURE(RECT, TEXTURE2D, GUICONTENT)

Declaration

```
public static Texture2D DrawTexture(Rect rect, Texture2D texture, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
texture	The input texture
content	The GUIContent for the field

Returns

The texture input into the inspector

Description

Draws a single texture field

DRAWTEXTUREWITHINT(TEXTURE2D, INT, GUICONTENT)

Declaration

```
public static (Texture2D, int) DrawTextureWithInt(Texture2D texture, int intValue, GUIContent content)
```

Parameters

Parameter	Description
texture	The input texture
intValue	The input integer value
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Int Value

Description

Draws a single texture field with an accompanied integer value

---

DRAWTEXTUREWITHINT(RECT, TEXTURE2D, INT, GUICONTENT)

Declaration

```
public static (Texture2D, int) DrawTextureWithInt(Rect rect, Texture2D texture, int intValue, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
texture	The input texture
intValue	The input integer value
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Int Value

Description

Draws a single texture field with an accompanied integer value

DRAWTEXTUREWITHINTSLIDER(TEXTURE2D, INT, INT, INT, GUICONTENT)

Declaration

```
public static (Texture2D, int) DrawTextureWithIntSlider(Texture2D texture, int sliderValue, int sliderMin, int sliderMax, GUIContent content)
```

Parameters

Parameter	Description
texture	The input texture
sliderValue	The input slider value
sliderMin	The minimum value that the slider will allow
sliderMax	The maximum value that the slider will allow
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Slider Value

Description

Draws a single texture field with an accompanied integer slider

DRAWTEXTUREWITHINTSLIDER(RECT, TEXTURE2D, INT, INT, INT, GUICONTENT)

Declaration

```
public static (Texture2D, int) DrawTextureWithIntSlider(Rect rect, Texture2D texture, int sliderValue, int sliderMin, int sliderMax, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
texture	The input texture
sliderValue	The input slider value
sliderMin	The minimum value that the slider will allow
sliderMax	The maximum value that the slider will allow
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Slider Value

Description

Draws a single texture field with an accompanied integer slider

DRAWTEXTUREWITHFLOAT(TEXTURE2D, FLOAT, GUICONTENT)

Declaration

```
public static (Texture2D, float) DrawTextureWithFloat(Texture2D texture, float floatValue, GUIContent content)
```

Parameters

Parameter	Description
texture	The input texture
floatValue	The input float value
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Float Value

Description

Draws a single texture field with an accompanied float value

DRAWTEXTUREWITHFLOAT(RECT, TEXTURE2D, FLOAT, GUICONTENT)

Declaration

```
public static (Texture2D, float) DrawTextureWithFloat(Rect rect, Texture2D texture, float floatValue, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
texture	The input texture
floatValue	The input float value
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Float Value

Description

Draws a single texture field with an accompanied float value

DRAWTEXTUREWITHSLIDER(TEXTURE2D, FLOAT, FLOAT, FLOAT, GUICONTENT)

Declaration

```
public static (Texture2D, float) DrawTextureWithSlider(Texture2D texture, float sliderValue, float sliderMin, float sliderMax, GUIContent content)
```

Parameters

Parameter	Description
texture	The input texture
sliderValue	The input slider value
sliderMin	The minimum value that the slider will allow
sliderMax	The maximum value that the slider will allow
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Slider Value

Description

Draws a single texture field with an accompanied slider

DRAWTEXTUREWITHSLIDER(RECT, TEXTURE2D, FLOAT, FLOAT, FLOAT, GUICONTENT)

Declaration

```
public static (Texture2D, float) DrawTextureWithSlider(Rect rect, Texture2D texture, float sliderValue, float sliderMin, float sliderMax, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
texture	The input texture
sliderValue	The input slider value
sliderMin	The minimum value that the slider will allow
sliderMax	The maximum value that the slider will allow
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Slider Value

Description

Draws a single texture field with an accompanied slider

DRAWTEXTUREWITHCOLOR(TEXTURE2D, COLOR, BOOL, GUICONTENT)

Declaration

```
public static (Texture2D, Color) DrawTextureWithColor(Texture2D texture, Color colorValue, bool hdr, GUIContent content)
```

Parameters

Parameter	Description
texture	The input texture
colorValue	The input color value
hdr	If the color is HDR
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Color Value

Description

Draws a single texture field with an accompanied color field

DRAWTEXTUREWITHCOLOR(RECT, TEXTURE2D, COLOR, BOOL, GUICONTENT)

Declaration

```
public static (Texture2D, Color) DrawTextureWithColor(Rect rect, Texture2D texture, Color colorValue, bool hdr, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
texture	The input texture
colorValue	The input color value
hdr	If the color is HDR
content	The GUIContent for the field

Returns

Item1: Texture, Item2: Color Value

Description

Draws a single texture field with an accompanied color field

DRAWTEXTURES(TEXTURE2D[], GUICONTENT[])

Declaration

```
public static Texture2D[] DrawTextures(Texture2D[] textures, GUIContent[] content = null)
```

Parameters

Parameter	Description
textures	The input textures that will be drawn in order
content	The GUIContent for each field in orderIf unassigned each texture will be named "Texture 1", "Texture 2", ...

Returns

Array of textures input into the inspector

Description

Draws multiple texture fields in sequence from the given texture array

---

DRAWVECTOR2FIELD(VECTOR2, GUICONTENT)

Declaration

```
public static Vector2 DrawVector2Field(Vector2 value, GUIContent content)
```

Parameters

Parameter	Description
value	The input Vector2 value
content	The GUIContent for the field

Returns

The Vector2 input into the inspector

Description

Draws a Vector2 field removing the space at the end which EditorGUI.Vector2Field draws

---

DRAWVECTOR2FIELD(RECT, VECTOR2, GUICONTENT)

Declaration

```
public static Vector2 DrawVector2Field(Rect rect, Vector2 value, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
value	The input Vector2 value
content	The GUIContent for the field

Returns

The Vector2 input into the inspector

Description

Draws a Vector2 field removing the space at the end which EditorGUI.Vector2Field draws

---

DRAWVECTOR2INTFIELD(VECTOR2INT, GUICONTENT)

Declaration

```
public static Vector2Int DrawVector2IntField(Vector2Int value, GUIContent content)
```

Parameters

Parameter	Description
value	The input Vector2Int value
content	The GUIContent for the field

Returns

The Vector2Int input into the inspector

Description

Draws a Vector2Int field removing the space at the end which EditorGUI.Vector2IntField draws

---

DRAWVECTOR2INTFIELD(RECT, VECTOR2INT, GUICONTENT)

Declaration

```
public static Vector2Int DrawVector2IntField(Rect rect, Vector2Int value, GUIContent content)
```

Parameters

Parameter	Description
rect	The space that the field will use
value	The input Vector2Int value
content	The GUIContent for the field

Returns

The Vector2Int input into the inspector

Description

Draws a Vector2Int field removing the space at the end which EditorGUI.Vector2IntField draws

---

DRAWTILINGOFFSET(MATERIALPROPERTY, STRING, STRING)

Declaration

```
public static void DrawTilingOffset(MaterialProperty tilingOffsetProperty, string tilingLabel = "Scale", string offsetLabel = "Offset")
```

Parameters

Parameter	Description
tilingOffsetProperty	A float4 property that will be used and updated for the tiling and offset values
tilingLabel	The label to show for the tiling field
offsetLabel	The label to show for the offset field

Description

Draws tiling offset fields using and updating values in a float4 materialProperty

Uses the DrawVector2Field function to remove empty space

DRAWTILINGOFFSET(VECTOR4, STRING, STRING)

Declaration

```
public static Vector4 DrawTilingOffset(Vector4 tilingOffset, string tilingLabel = "Tiling", string offsetLabel = "Offset")
```

Parameters

Parameter	Description
tilingOffset	The initial tiling offset value
tilingLabel	The label to show for the tiling field
offsetLabel	The label to show for the offset field

Returns

The new tiling offset value

Description

Draws tiling offset fields using and updating values in a float4 materialProperty

Uses the DrawVector2Field function to remove empty space
